

Step-7 (Track A) — Horizon x Cov-Duration Bake-off Master Dashboard (Sigma 30s / 45s / 60s, MASKED)

2022-2025 · cov-dur x horizon(nextwin..1h) x metric(LE/AI) x 15 learners · fixed Step-6 predictor [rv, s\_slow, s\_fast, v]; realized-VOL target aggregated to horizon · AI headline / LE diagnostic · next-window COINCIDENT, no next-day forecast (deferred Track B) · cells finished: 16 · durations present: 60s

**What this run is.** A **cov-duration x prediction-horizon** bake-off. No new covariance is built: the 30/45/60 s **masked Sigma state** (Steps 3-6) is unchanged; Step 7 moves ONE axis - the target's forward SPAN H (nextwin, 2m, 5m, 10m, 15m, 30m, 1h). For every (cov-dur x horizon x metric) cell we fit the COMPLETE 15-learner panel (GridSearch + purged walk-forward) with the **fixed Step-6 predictor** [rv, s\_slow, s\_fast, v] and a realized-VOL target aggregated to H. **LE** features = Log-Euclidean Step-6 dial; **AI** features = affine-invariant dial (PC1 of the AI whitened tangent) + AI geodesic velocity **AI** **LE** . **Every number below is labelled with its cov-duration AND its horizon.**

**The claim being tested.** Holding the covariance state fixed: (1) at which HORIZON does the dial stop being coincident and start adding OOS value over persistence (the **dR2(H)** curve); (2) does a FLEXIBLE learner beat linear at that horizon (capacity); (3) is the answer METRIC-ROBUST (AI vs LE)? **next-window is COINCIDENT** (persistence wins); longer horizons test whether the dial becomes LEADING. There is NO next-day forecast here - that is the deferred Track B.

60s  
cov durations

nextwin . 30s . 45s . 60s . 2m . 5m . 10m . 15m . 30m . 1h  
horizons

LE / AI  
metrics

15  
learners / cell

16 / 20  
cells finished

4  
cells N/A (skipped)

60s  
durations present

**Coverage grid (cov-duration x horizon).** which cells the detached run has finished (LE+AI green / one metric amber), which are physically SKIPPED (N/A = horizon finer than the cov window), and which are still pending (—).

| Sigma \ horizon | nextwin | 30s | 45s | 60s   | 2m    | 5m    | 10m   | 15m   | 30m   | 1h    |
|-----------------|---------|-----|-----|-------|-------|-------|-------|-------|-------|-------|
| 60s             | LE+AI   | N/A | N/A | LE+AI | LE+AI | LE+AI | LE+AI | LE+AI | LE+AI | LE+AI |

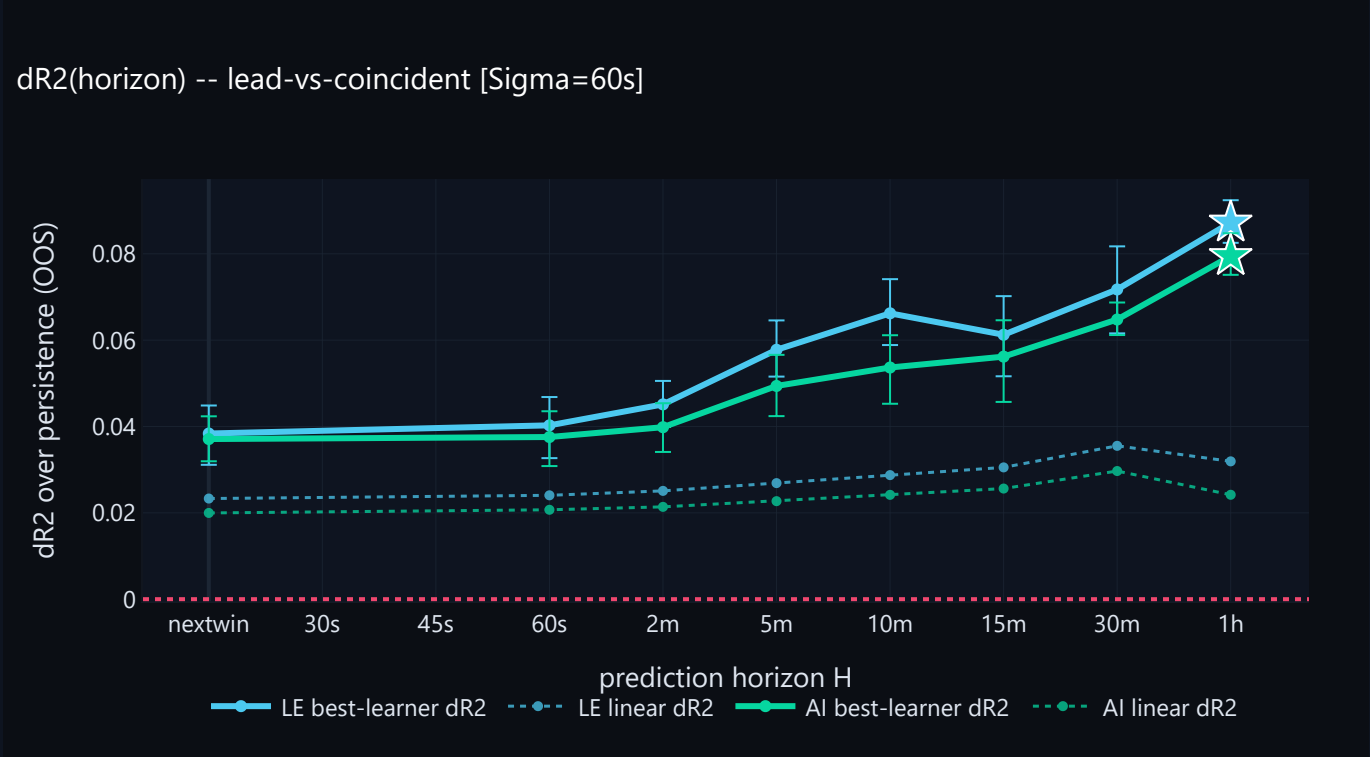
**Peak lead-vs-coincident edge (per cov-dur x metric).** the horizon where the best learner's dR2 over persistence is largest, and whether its 95% CI clears 0.

| Sigma | metric | horizons done | peak dR2 horizon | peak dR2 | sig (CI>0) |
|-------|--------|---------------|------------------|----------|------------|
| 60s   | LE     | 8 / 10        | 1h               | 0.087    | True       |
| 60s   | AI     | 8 / 10        | 1h               | 0.079    | True       |

**THE headline of Track A - lead-vs-coincident.** For each cov-duration, plot the dial's OOS edge over persistence **dR2(horizon)** as the target span H grows (x = nextwin, 2m, 5m, 10m, 15m, 30m, 1h). Solid = the BEST learner's dR2; dotted = the linear learner's dR2. LE (blue) vs AI (green) **AI** **LE** . **The question:** at next-window the dial is COINCIDENT (persistence wins, dR2 small); does dR2 GROW with horizon (dial becomes LEADING)? The marked star = the horizon that PEAKS dR2 (the characteristic horizon).

dR2(horizon) per cov-duration — native 30s / 45s / 60s Sigma; LE vs AI

Sigma=60s dR2(horizon) lead-vs-coincident



**In plain terms** How much extra vol-forecast accuracy the covariance dial adds over pure persistence, at each forward span.

**Math / formula**  $dR2(H) = R2\_OOS(\text{best learner}) - R2\_OOS(\text{persistence=rv-only})$ ; block-bootstrap 95% CI over days.

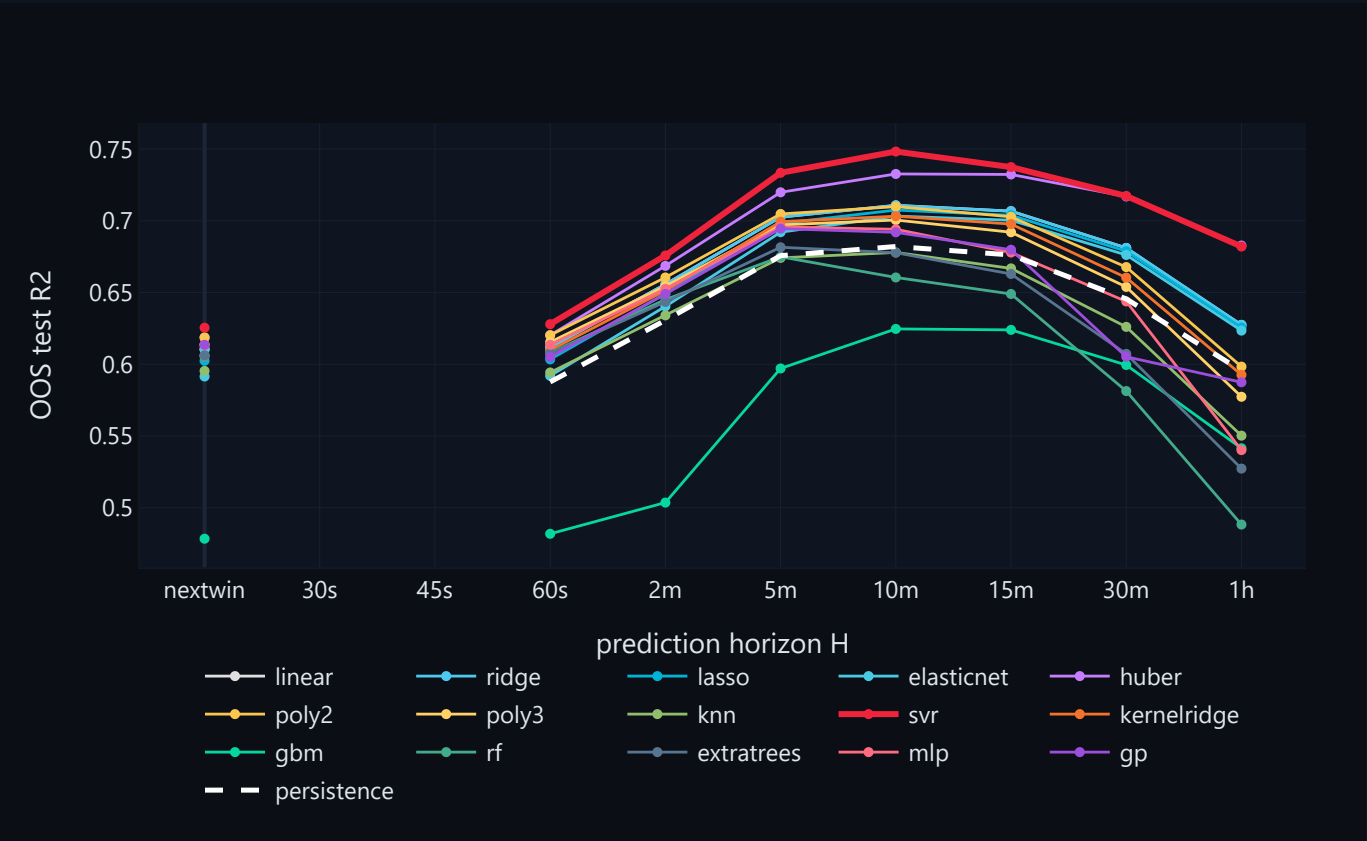
**How to read it** flat/low near nextwin = coincident; rising with H = the dial becomes leading; star marks the peak H.

**Insight** LE: peak dR2 **0.087** @ 1h (nextwin 0.038) [sig] | AI: peak dR2 **0.079** @ 1h (nextwin 0.037) [sig]

**Performance-comparison plots.** The same numbers as the tables, drawn for the eye. Every chart is labelled with its cov-duration / horizon / metric; horizons are always in canonical order (nextwin, 30s, 45s, 60s, 2m, 5m, 10m, 15m, 30m, 1h); skipped cells (horizon finer than the cov window) are greyed/omitted. Colors = learners (svr = the usual winner, highlighted).

1 · OOS test R2 vs horizon — 15 learners + persistence — one chart per cov-duration x metric

Sigma=60s OOS R2 vs horizon [Sigma=60s | metric=LE] LE



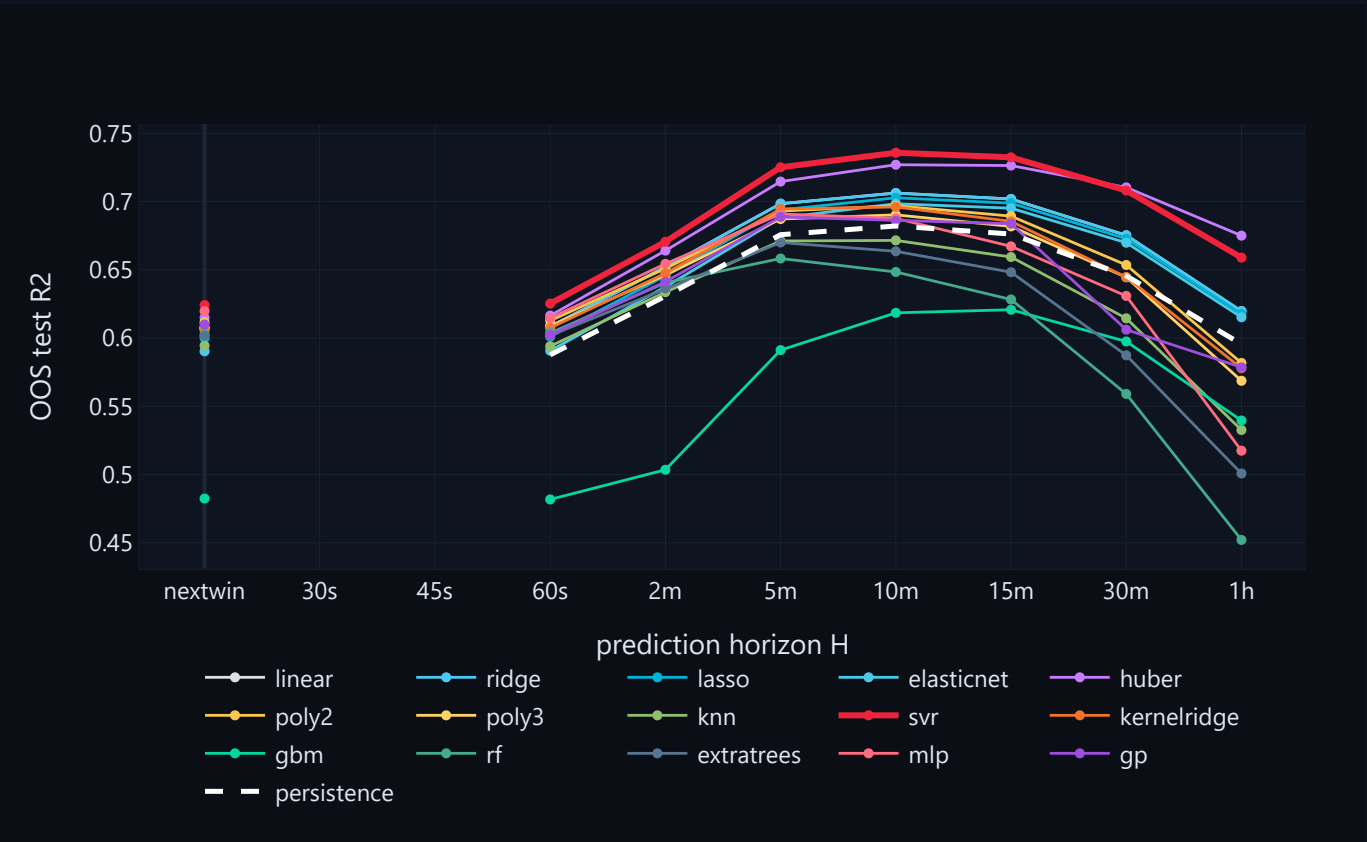
**In plain terms** How each of the 15 learners tracks vol as the forecast span grows, vs the persistence floor.

**Math / formula**  $y = R2\_OOS(model)$  per horizon; dashed white = persistence (rv-only) baseline.

**How to read it** lines above the dashed floor beat persistence; watch the interior peak (~10m).

**Insight** cov 60s, metric LE; missing points = unfinished/skipped horizons.

Sigma=60s OOS R2 vs horizon [Sigma=60s | metric=AI] AI



**In plain terms** How each of the 15 learners tracks vol as the forecast span grows, vs the persistence floor.

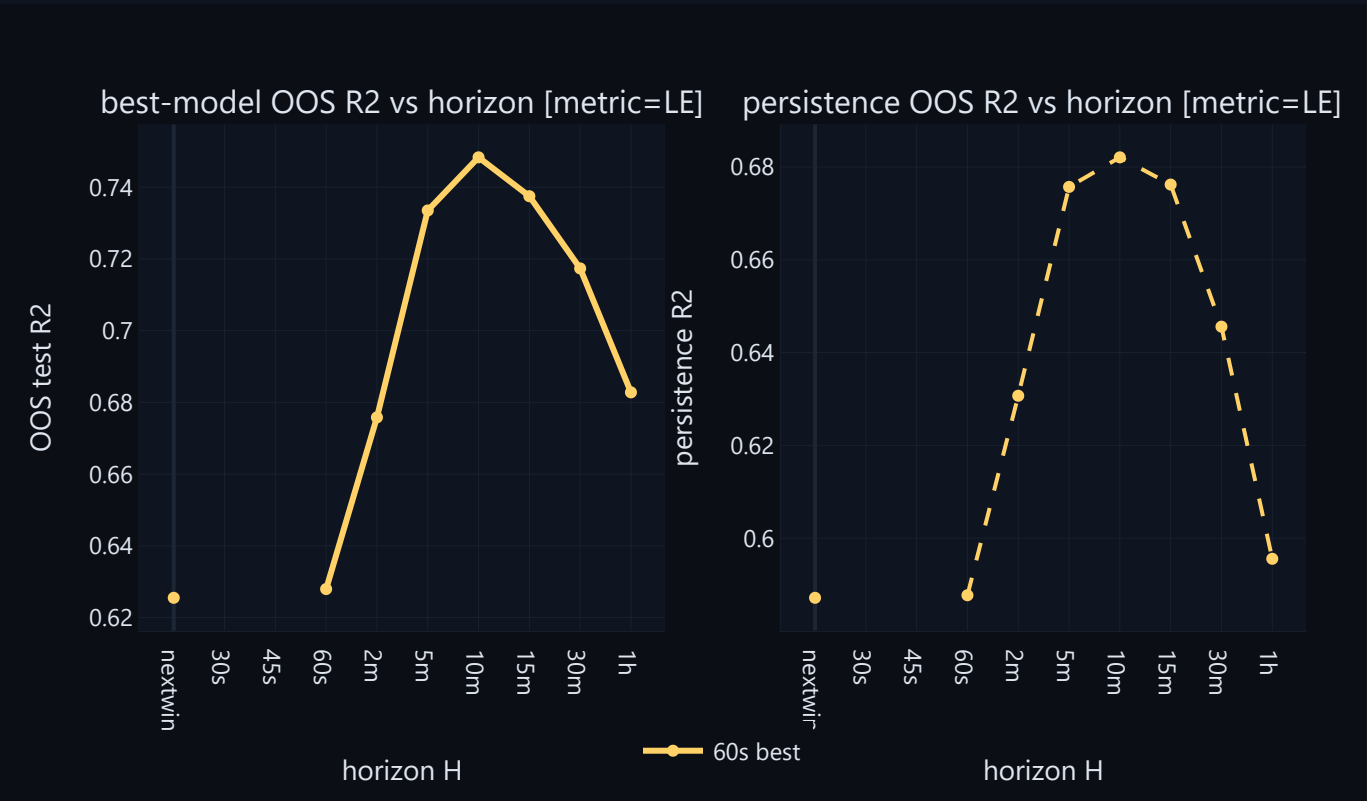
**Math / formula**  $y = R2\_OOS(model)$  per horizon; dashed white = persistence (rv-only) baseline.

**How to read it** lines above the dashed floor beat persistence; watch the interior peak (~10m).

**Insight** cov 60s, metric AI; missing points = unfinished/skipped horizons.

3 · Cross-duration overlay — 30s vs 45s vs 60s — best-model R2 (left) and persistence R2 (right) vs horizon

Sigma=60s Cross-duration: best-model & persistence R2 vs horizon [metric=LE] LE



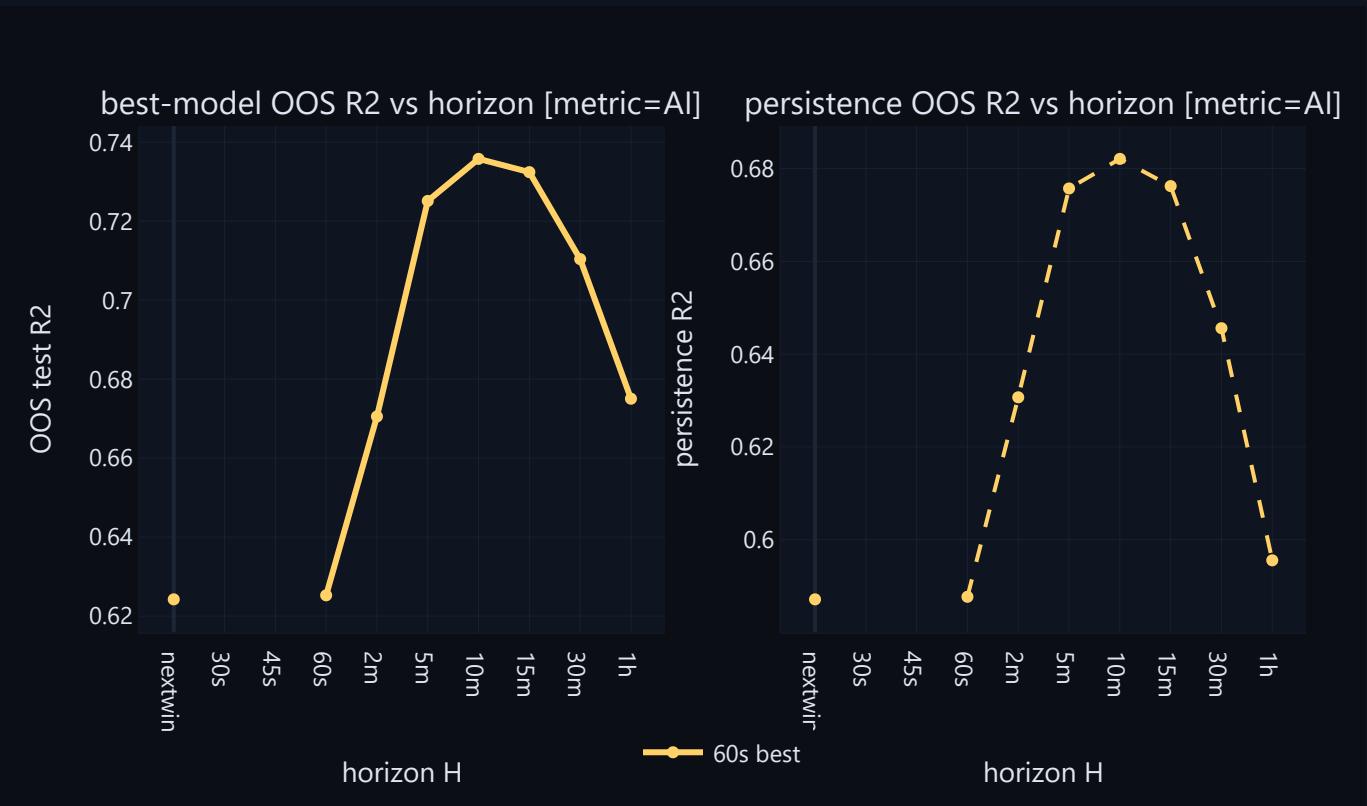
**In plain terms** Does a coarser covariance window (60s) give a higher absolute R2 than a finer one (30s)?

**Math / formula** left: best-learner R2 per horizon for 30/45/60s; right: the persistence floor per horizon.

How to read it  
Insight

curves ordered 60s>45s>30s = coarser Sigma is more predictable (both model and floor rise).  
metric LE; both panels share the horizon axis.

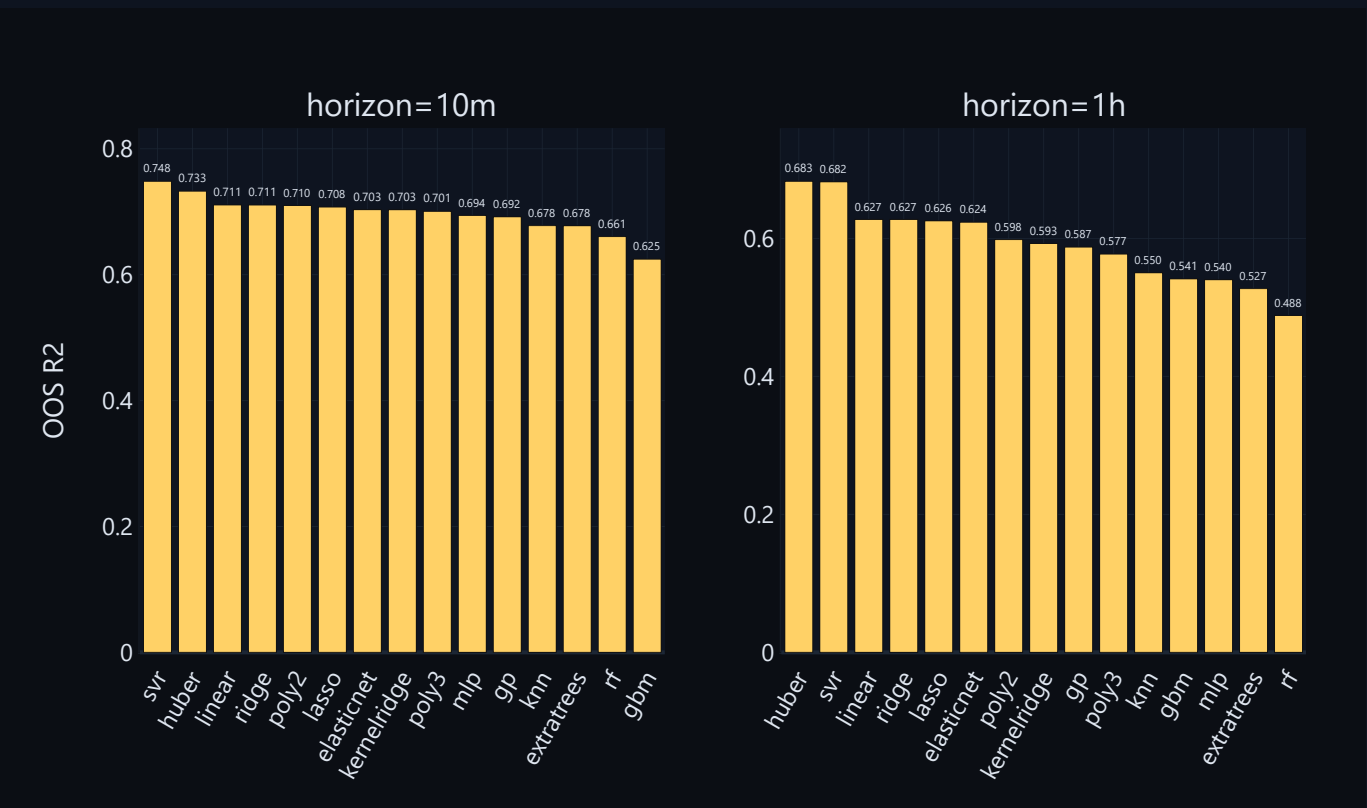
Sigma=60s Cross-duration: best-model & persistence R2 vs horizon [metric=AI] AI



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Math / formula left: best-learner R2 per horizon for 30/45/60s; right: the persistence floor per horizon.  
How to read it curves ordered 60s>45s>30s = coarser Sigma is more predictable (both model and floor rise).  
Insight metric AI; both panels share the horizon axis.

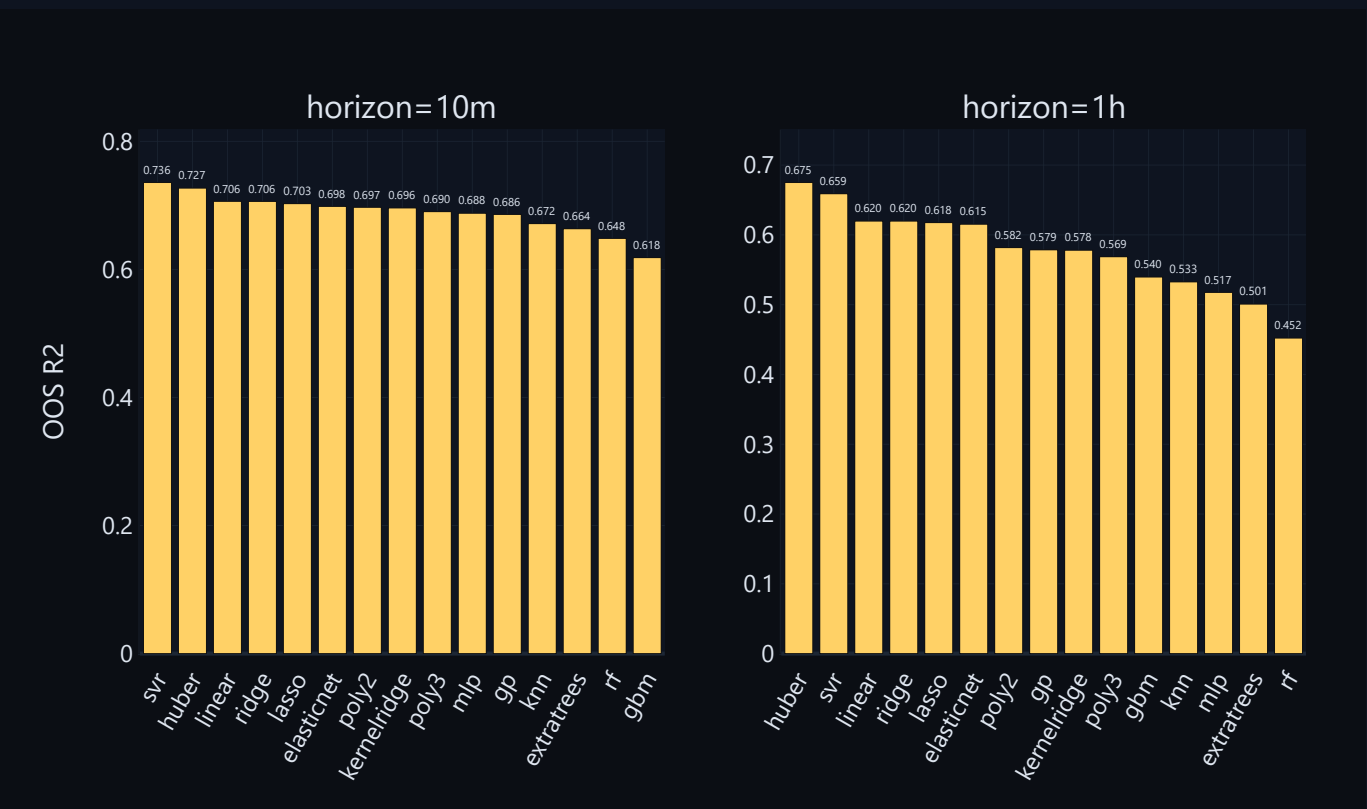
4 · Model ranking (OOS R2, sorted) — at 10m and 1h — per cov-duration x metric; winner highlighted

Sigma=60s Learner ranking [Sigma=60s | metric=LE] LE



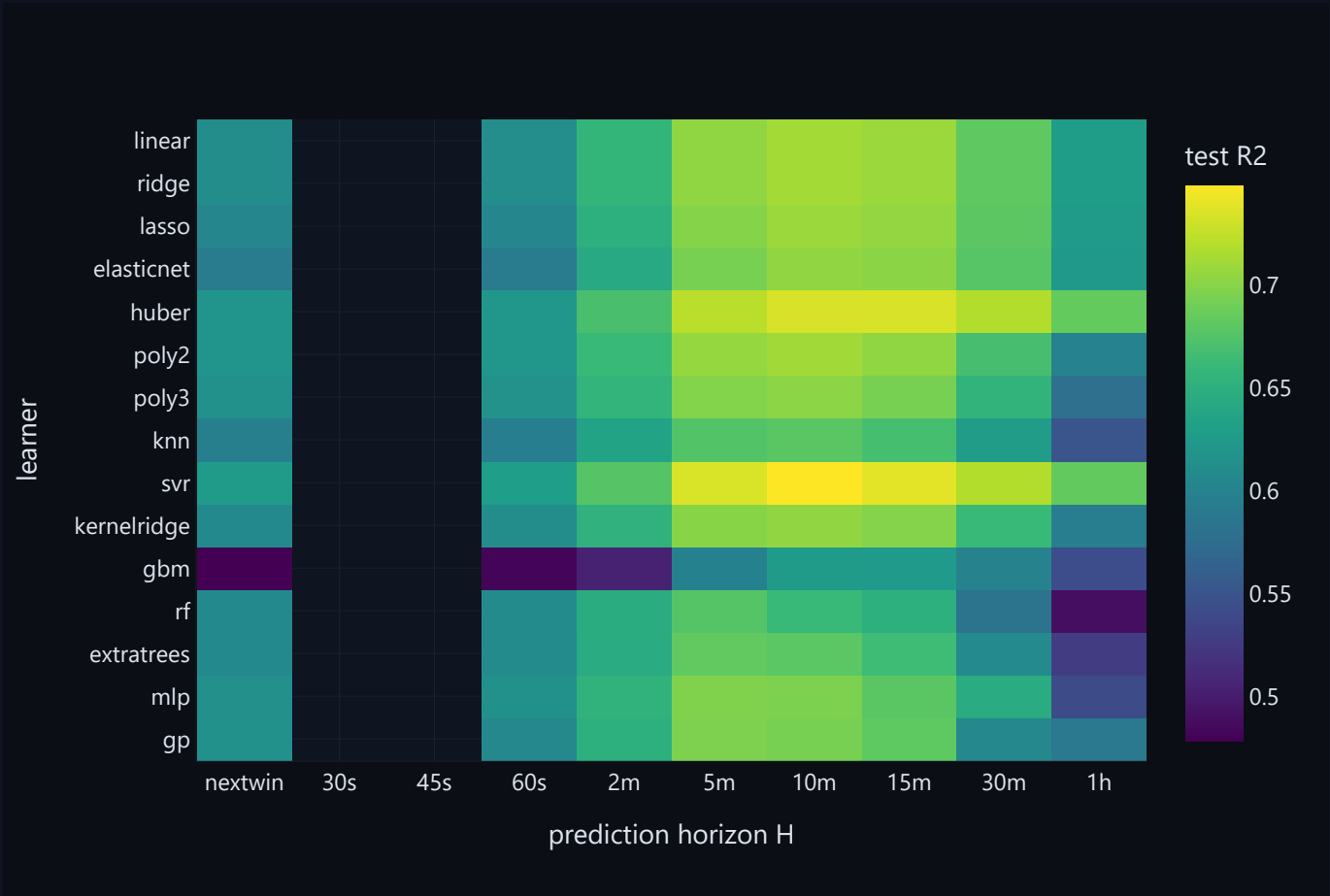
In plain terms Which learner wins at the interesting horizons, and by how much.  
Math / formula bars = R2\_OOS per learner, sorted descending; amber = svr (the usual winner).  
How to read it a tall gap from #1 to linear = capacity helps; a flat top = many models tie.  
Insight cov 60s, metric LE, horizons 10m, 1h.

Sigma=60s Learner ranking [Sigma=60s | metric=AI] AI



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Insight cov 60s, metric AI, horizons 10m, 1h.

Sigma=60s OOS R2 heatmap [Sigma=60s | metric=LE] LE



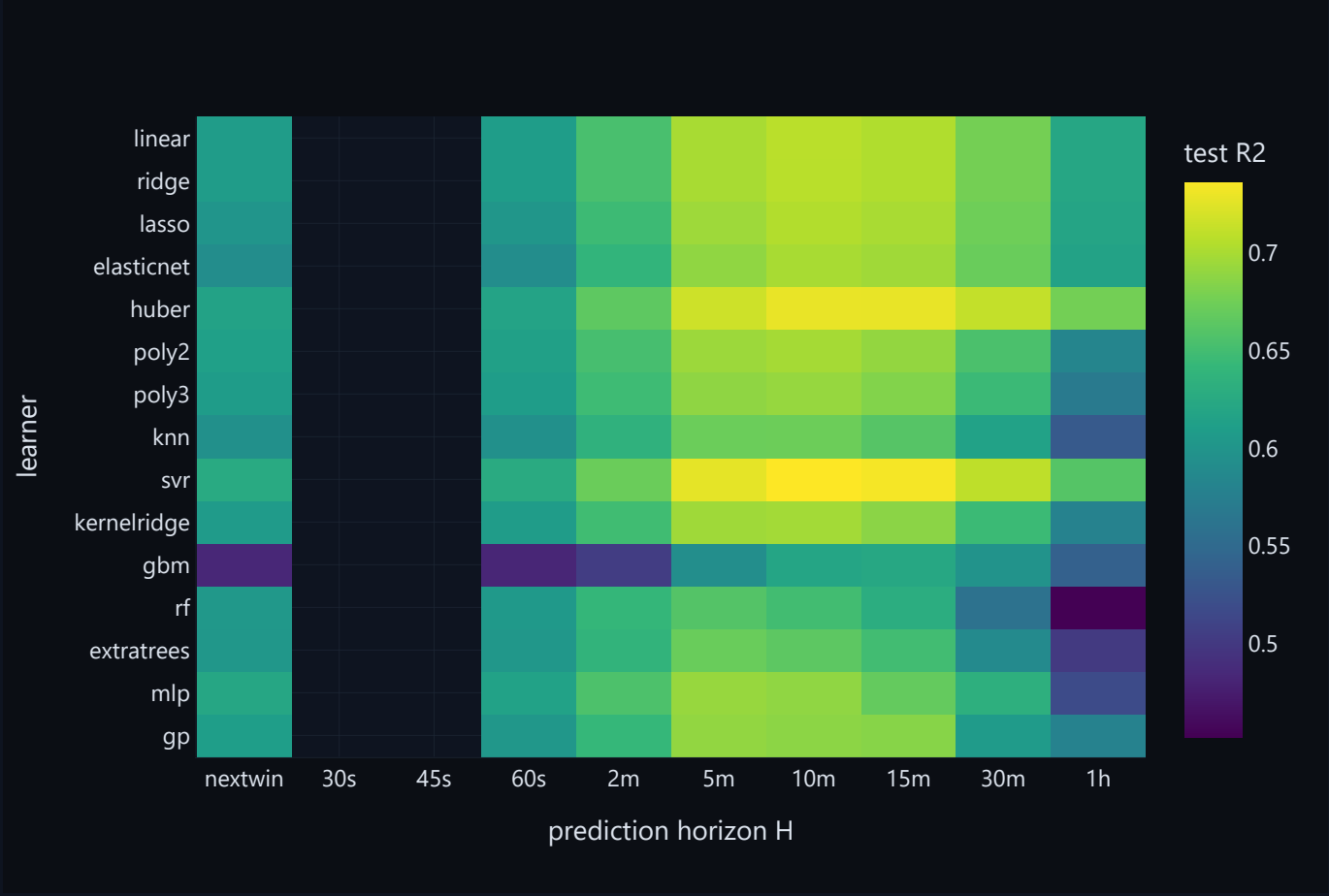
**In plain terms** The whole learner x horizon accuracy grid at a glance.

**Math / formula** cell colour =  $R2\_OOS(learner, horizon)$ ; grey = skipped (horizon < cov window).

**How to read it** bright bands = strong learner/horizon combos; a bright column = an easy horizon.

**Insight** cov 60s, metric LE; 15 learners x 8 finished horizons.

Sigma=60s OOS R2 heatmap [Sigma=60s | metric=AI] AI



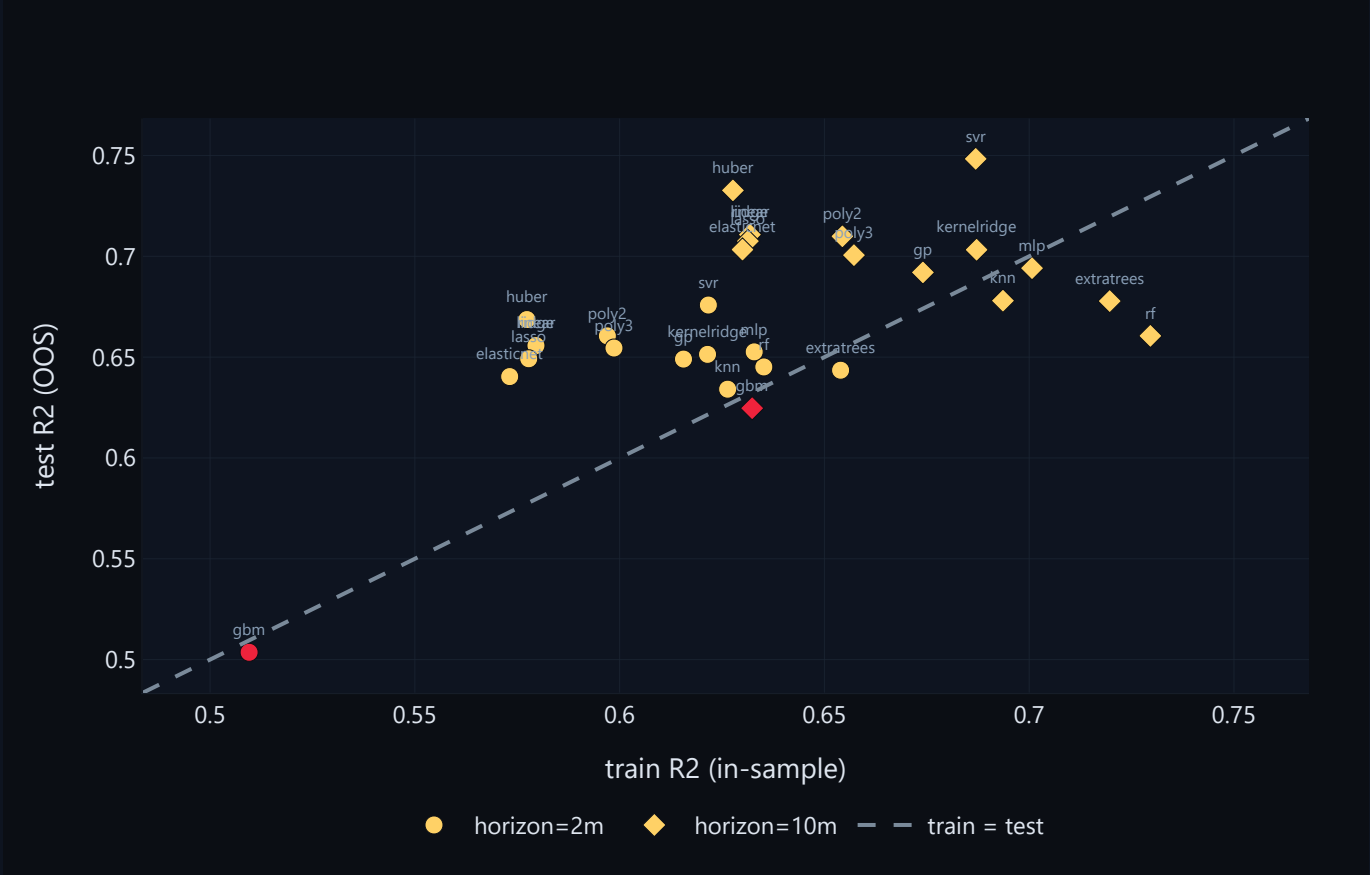
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**Insight** cov 60s, metric AI; 15 learners x 8 finished horizons.

Sigma=60s Overfit: train vs test R2 [Sigma=60s | metric=LE] LE



In plain terms

Math / formula

How to read it

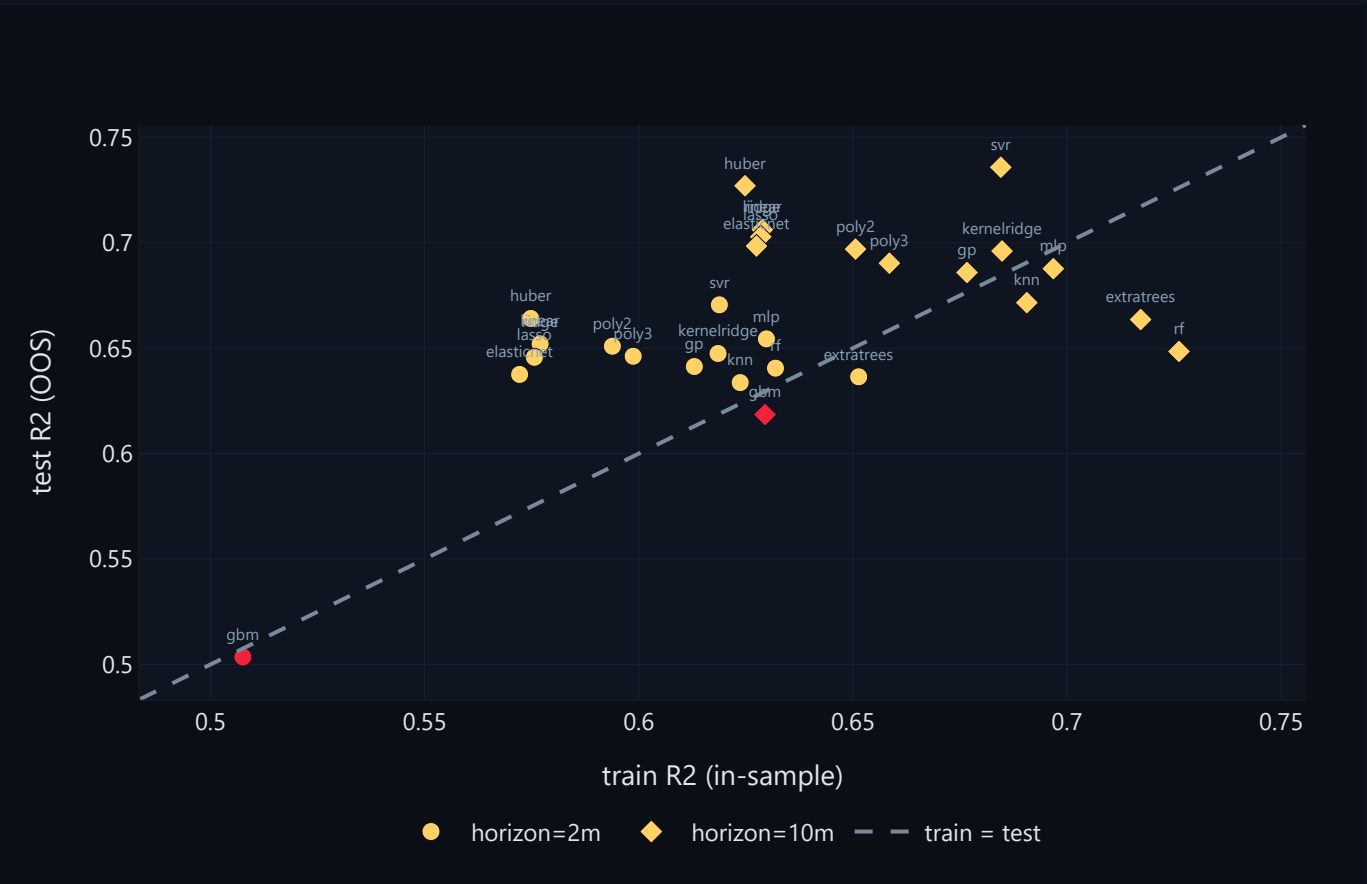
Insight

How much each learner's in-sample fit exceeds its out-of-sample accuracy (the gap = overfit).  
x = train R2, y = test R2; dashed = no gap; distance BELOW the line = overfit.  
points far under the diagonal (watch gbm, red) overfit; on/near it generalise.  
cov 60s, metric LE, horizons 2m, 10m.

Sigma=60s

Overfit: train vs test R2 [Sigma=60s | metric=AI]

AI



In plain terms

Math / formula

How to read it

Insight

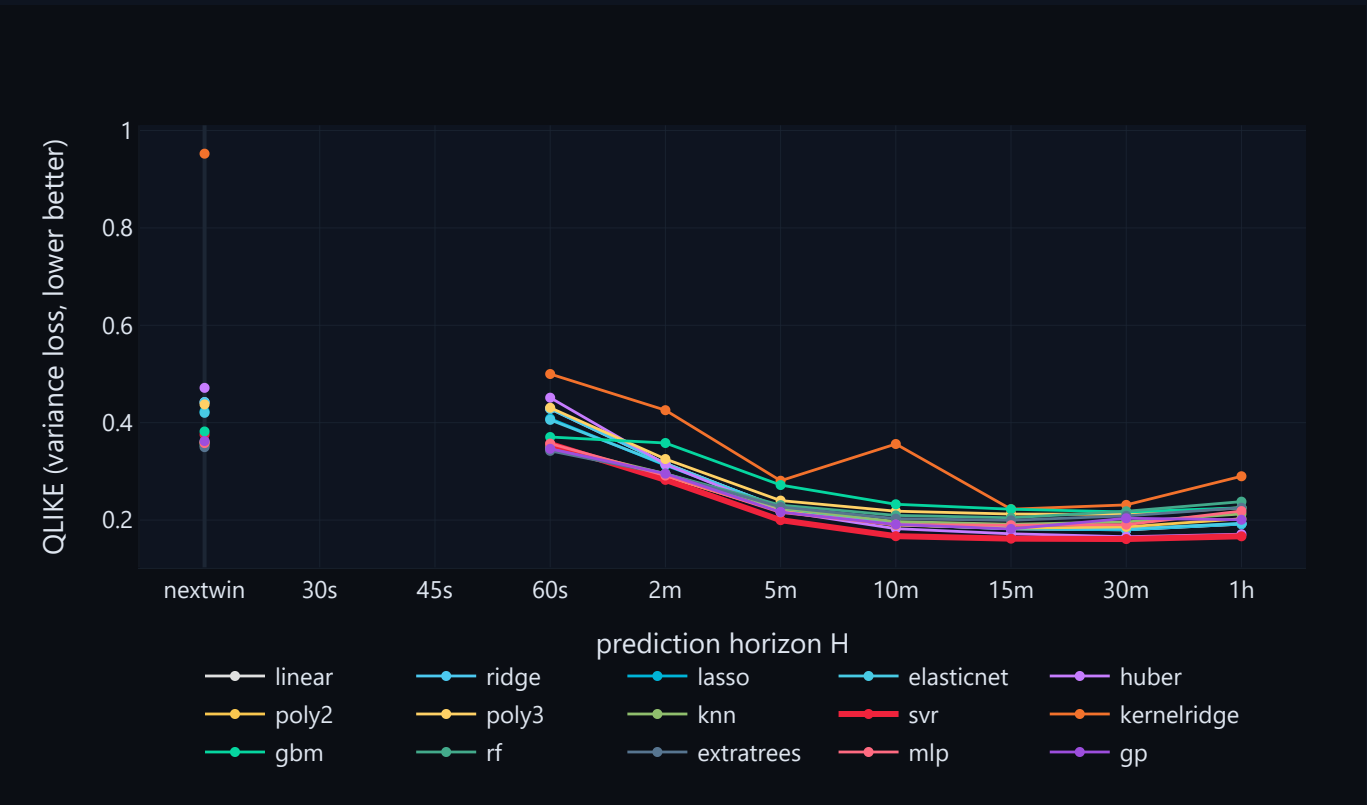
How much each learner's in-sample fit exceeds its out-of-sample accuracy (the gap = overfit).  
x = train R2, y = test R2; dashed = no gap; distance BELOW the line = overfit.  
points far under the diagonal (watch gbm, red) overfit; on/near it generalise.  
cov 60s, metric AI, horizons 2m, 10m.

8 · QLIKE vs horizon — 15 learners (lower is better) — per cov-duration x metric

Sigma=60s

QLIKE vs horizon [Sigma=60s | metric=LE]

LE



In plain terms

Math / formula

How to read it

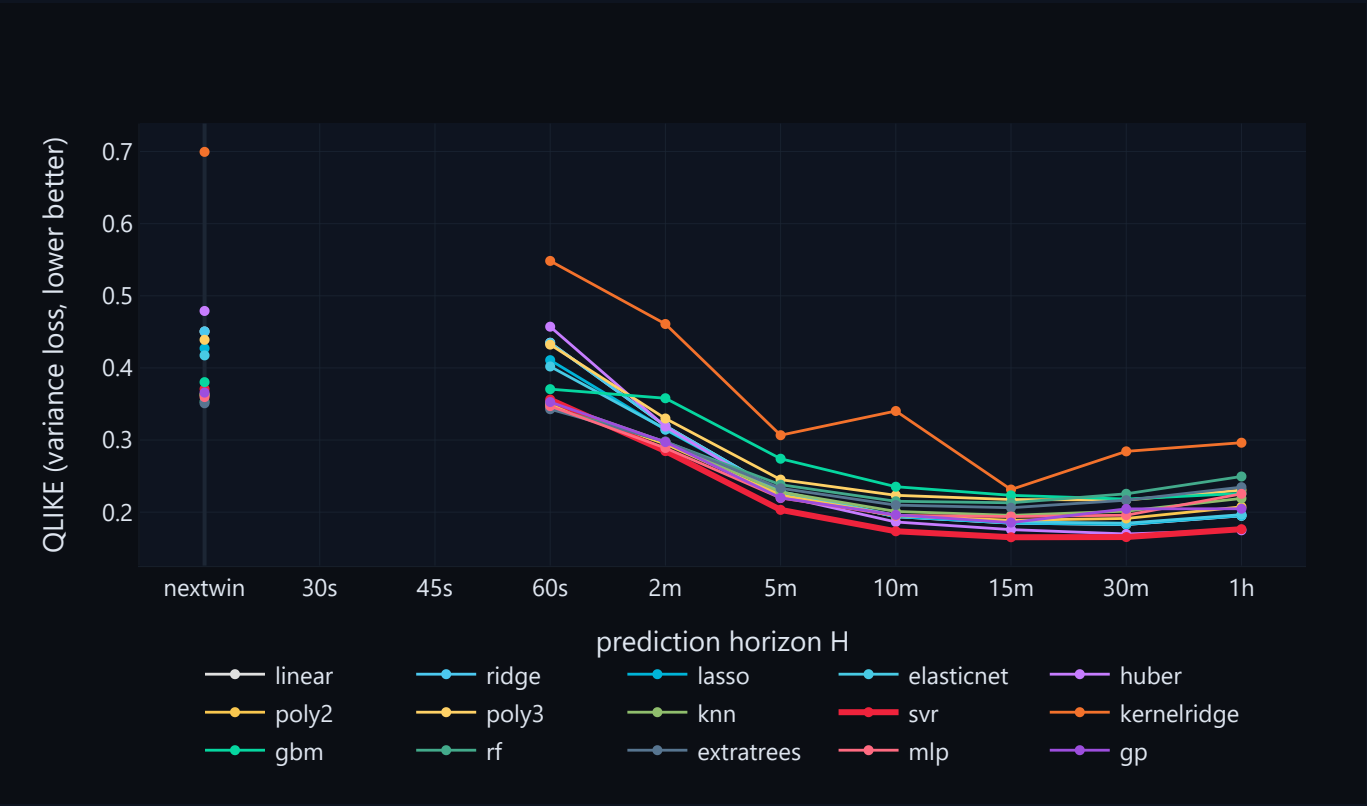
Insight

Does the accuracy story survive the stricter QLIKE variance loss (not just R2)?  
y = QLIKE\_OOS(model) per horizon = mean(r - ln r - 1), r = variance ratio; lower = better.  
lines that stay low across horizons are robust; spikes flag loss-sensitive learners.  
cov 60s, metric LE.

Sigma=60s

QLIKE vs horizon [Sigma=60s | metric=AI]

AI



In plain terms

Does the accuracy story survive the stricter QLIKE variance loss (not just R2)?

Math / formula  
How to read it  
Insight

y = QLIKE\_OOS(model) per horizon = mean(r - ln r - 1), r = variance ratio; lower = better.  
lines that stay low across horizons are robust; spikes flag loss-sensitive learners.  
cov 60s, metric AI.

**The full matrix.** For EVERY (cov-dur x horizon x metric) cell, all 15 learners x {train R2, test R2, gap, dR2 [CI95], sig, RMSE, MAE, QLIKE, bias}. Grouped by cov-duration; within each, a labelled block per horizon with LE and AI side by side. Every table title carries **Sigma=<dur>s | horizon=<H> | metric=<LE/AI>**. The star row = best-by-test-R2. RMSE/MAE/bias in vol units; QLIKE on variance.

Sigma=60s cov-duration 60s — 10 horizon(s) finished

horizon=nextwin Sigma=60s · horizon=nextwin (H=60s) — metrics: LE + AI

LE Sigma=60s | horizon=nextwin | metric=LE

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.542    | 0.610   | -0.068 | 0.023 [0.021, 0.025]    | yes | 2.54e-04 | 1.98e-04 | 6.44e-08 | 0.442    | 5.92e-05 | 3.03e-04 | 2.29e-04 | 9.19e-08 | 0.518    | 9.45e-17  |
| ridge       | 0.542    | 0.610   | -0.068 | 0.023 [0.021, 0.025]    | yes | 2.54e-04 | 1.98e-04 | 6.44e-08 | 0.442    | 5.92e-05 | 3.03e-04 | 2.29e-04 | 9.19e-08 | 0.518    | -5.46e-18 |
| lasso       | 0.540    | 0.602   | -0.063 | 0.015 [0.014, 0.017]    | yes | 2.56e-04 | 2.00e-04 | 6.57e-08 | 0.423    | 6.22e-05 | 3.04e-04 | 2.29e-04 | 9.24e-08 | 0.493    | 4.29e-20  |
| elasticnet  | 0.533    | 0.592   | -0.059 | 0.004 [0.003, 0.006]    | yes | 2.60e-04 | 2.03e-04 | 6.75e-08 | 0.420    | 6.56e-05 | 3.06e-04 | 2.31e-04 | 9.38e-08 | 0.486    | 8.43e-21  |
| huber       | 0.540    | 0.619   | -0.078 | 0.032 [0.030, 0.033]    | yes | 2.51e-04 | 1.96e-04 | 6.30e-08 | 0.472    | 4.45e-05 | 3.04e-04 | 2.28e-04 | 9.22e-08 | 0.555    | -1.57e-05 |
| poly2       | 0.558    | 0.619   | -0.061 | 0.031 [0.028, 0.035]    | yes | 2.51e-04 | 1.96e-04 | 6.30e-08 | 0.359    | 7.15e-05 | 2.98e-04 | 2.24e-04 | 8.88e-08 | 0.425    | 1.68e-15  |
| poly3       | 0.558    | 0.614   | -0.056 | 0.027 [0.024, 0.029]    | yes | 2.53e-04 | 1.98e-04 | 6.38e-08 | 0.437    | 7.39e-05 | 2.98e-04 | 2.25e-04 | 8.87e-08 | 0.434    | 2.39e-13  |
| knn         | 0.582    | 0.595   | -0.013 | 0.008 [0.002, 0.015]    | yes | 2.59e-04 | 2.03e-04 | 6.69e-08 | 0.361    | 8.36e-05 | 2.90e-04 | 2.17e-04 | 8.39e-08 | 0.392    | -3.65e-06 |
| svr ★       | 0.579    | 0.626   | -0.046 | 0.038 [0.031, 0.045]    | yes | 2.49e-04 | 1.94e-04 | 6.19e-08 | 0.375    | 6.55e-05 | 2.91e-04 | 2.17e-04 | 8.44e-08 | 0.416    | -1.55e-05 |
| kernelridge | 0.577    | 0.606   | -0.028 | 0.018 [0.004, 0.030]    | yes | 2.55e-04 | 1.98e-04 | 6.52e-08 | 0.952    | 7.72e-05 | 2.91e-04 | 2.18e-04 | 8.49e-08 | 0.802    | -1.77e-06 |
| gbm         | 0.474    | 0.478   | -0.004 | -0.109 [-0.117, -0.098] | no  | 2.94e-04 | 2.31e-04 | 8.62e-08 | 0.382    | 1.21e-04 | 3.25e-04 | 2.45e-04 | 1.06e-07 | 0.372    | -2.12e-07 |
| rf          | 0.592    | 0.606   | -0.014 | 0.019 [0.013, 0.024]    | yes | 2.55e-04 | 2.01e-04 | 6.51e-08 | 0.353    | 8.68e-05 | 2.86e-04 | 2.15e-04 | 8.18e-08 | 0.376    | 6.51e-08  |
| extratrees  | 0.611    | 0.606   | 0.005  | 0.019 [0.014, 0.023]    | yes | 2.55e-04 | 2.01e-04 | 6.51e-08 | 0.350    | 8.75e-05 | 2.79e-04 | 2.09e-04 | 7.81e-08 | 0.348    | 8.43e-21  |
| mlp         | 0.589    | 0.613   | -0.024 | 0.026 [0.020, 0.031]    | yes | 2.53e-04 | 1.99e-04 | 6.39e-08 | 0.358    | 8.38e-05 | 2.87e-04 | 2.16e-04 | 8.25e-08 | 0.379    | 2.12e-06  |
| gp          | 0.570    | 0.614   | -0.044 | 0.026 [0.019, 0.033]    | yes | 2.53e-04 | 1.97e-04 | 6.39e-08 | 0.362    | 7.79e-05 | 2.94e-04 | 2.20e-04 | 8.63e-08 | 0.417    | -5.77e-06 |

best-by-test-R2 = svr (0.626) · linear 0.610 · persistence 0.587 · H=60s · 1801.4s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=60s | horizon=nextwin | metric=AI

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.540    | 0.607   | -0.067 | 0.020 [0.018, 0.022]    | yes | 2.55e-04 | 1.99e-04 | 6.49e-08 | 0.451    | 5.76e-05 | 3.04e-04 | 2.30e-04 | 9.24e-08 | 0.523    | -3.25e-16 |
| ridge       | 0.540    | 0.607   | -0.067 | 0.020 [0.018, 0.022]    | yes | 2.55e-04 | 1.99e-04 | 6.49e-08 | 0.451    | 5.76e-05 | 3.04e-04 | 2.30e-04 | 9.24e-08 | 0.523    | -5.10e-18 |
| lasso       | 0.538    | 0.600   | -0.062 | 0.013 [0.011, 0.015]    | yes | 2.57e-04 | 2.01e-04 | 6.61e-08 | 0.427    | 6.23e-05 | 3.05e-04 | 2.30e-04 | 9.28e-08 | 0.495    | 2.81e-21  |
| elasticnet  | 0.533    | 0.590   | -0.058 | 0.003 [0.002, 0.005]    | yes | 2.60e-04 | 2.03e-04 | 6.77e-08 | 0.417    | 6.72e-05 | 3.06e-04 | 2.31e-04 | 9.38e-08 | 0.482    | 2.81e-21  |
| huber       | 0.538    | 0.615   | -0.077 | 0.028 [0.026, 0.029]    | yes | 2.52e-04 | 1.96e-04 | 6.36e-08 | 0.479    | 4.31e-05 | 3.05e-04 | 2.29e-04 | 9.27e-08 | 0.562    | -1.58e-05 |
| poly2       | 0.554    | 0.612   | -0.058 | 0.024 [0.021, 0.028]    | yes | 2.53e-04 | 1.98e-04 | 6.42e-08 | 0.363    | 7.20e-05 | 2.99e-04 | 2.25e-04 | 8.95e-08 | 0.421    | -4.68e-16 |
| poly3       | 0.557    | 0.608   | -0.051 | 0.020 [0.016, 0.024]    | yes | 2.55e-04 | 1.99e-04 | 6.48e-08 | 0.439    | 7.63e-05 | 2.98e-04 | 2.25e-04 | 8.89e-08 | 0.427    | -1.20e-14 |
| knn         | 0.579    | 0.594   | -0.015 | 0.007 [0.002, 0.012]    | yes | 2.59e-04 | 2.03e-04 | 6.70e-08 | 0.362    | 8.37e-05 | 2.91e-04 | 2.18e-04 | 8.45e-08 | 0.394    | -4.26e-06 |
| svr ★       | 0.577    | 0.624   | -0.047 | 0.037 [0.032, 0.042]    | yes | 2.49e-04 | 1.94e-04 | 6.21e-08 | 0.371    | 6.48e-05 | 2.91e-04 | 2.18e-04 | 8.49e-08 | 0.415    | -1.48e-05 |
| kernelridge | 0.575    | 0.606   | -0.031 | 0.019 [0.010, 0.027]    | yes | 2.55e-04 | 1.99e-04 | 6.51e-08 | 0.700    | 7.76e-05 | 2.92e-04 | 2.18e-04 | 8.53e-08 | 0.871    | -1.44e-06 |
| gbm         | 0.473    | 0.482   | -0.009 | -0.105 [-0.113, -0.095] | no  | 2.92e-04 | 2.30e-04 | 8.55e-08 | 0.380    | 1.19e-04 | 3.25e-04 | 2.45e-04 | 1.06e-07 | 0.373    | -2.15e-07 |
| rf          | 0.589    | 0.603   | -0.014 | 0.016 [0.009, 0.021]    | yes | 2.56e-04 | 2.02e-04 | 6.57e-08 | 0.353    | 8.70e-05 | 2.87e-04 | 2.16e-04 | 8.25e-08 | 0.376    | 6.02e-08  |
| extratrees  | 0.608    | 0.602   | 0.007  | 0.015 [0.009, 0.019]    | yes | 2.57e-04 | 2.02e-04 | 6.58e-08 | 0.351    | 8.89e-05 | 2.80e-04 | 2.10e-04 | 7.86e-08 | 0.348    | -3.78e-21 |
| mlp         | 0.586    | 0.620   | -0.034 | 0.033 [0.027, 0.037]    | yes | 2.51e-04 | 1.97e-04 | 6.28e-08 | 0.360    | 7.65e-05 | 2.88e-04 | 2.17e-04 | 8.32e-08 | 0.392    | -6.26e-07 |
| gp          | 0.567    | 0.610   | -0.042 | 0.023 [0.014, 0.030]    | yes | 2.54e-04 | 1.98e-04 | 6.45e-08 | 0.366    | 7.61e-05 | 2.95e-04 | 2.20e-04 | 8.68e-08 | 0.415    | -5.71e-06 |

best-by-test-R2 = svr (0.624) · linear 0.607 · persistence 0.587 · H=60s · 1840.1s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=30s Sigma=60s · horizon=30s (H=30s) — metrics: LE + AI

LE Sigma=60s | horizon=30s | metric=LE

N/A — horizon < cov window (H=30s ≤ Sigma=60s). horizon 30s < cov 60s (cover 0.0%)

AI Sigma=60s | horizon=30s | metric=AI

N/A — horizon < cov window (H=30s ≤ Sigma=60s). horizon 30s < cov 60s (cover 0.0%)

horizon=45s Sigma=60s · horizon=45s (H=45s) — metrics: LE + AI

LE Sigma=60s | horizon=45s | metric=LE

N/A — horizon < cov window (H=45s ≤ Sigma=60s). horizon 45s < cov 60s (cover 0.0%)

AI Sigma=60s | horizon=45s | metric=AI

N/A — horizon < cov window (H=45s ≤ Sigma=60s). horizon 45s < cov 60s (cover 0.0%)

horizon=60s Sigma=60s · horizon=60s (H=60s) — metrics: LE + AI

LE Sigma=60s | horizon=60s | metric=LE

| learner    | train R2 | test R2 | gap    | dR2 [CI95]           | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|------------|----------|---------|--------|----------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear     | 0.544    | 0.612   | -0.067 | 0.024 [0.022, 0.026] | yes | 2.46e-04 | 1.94e-04 | 6.04e-08 | 0.429    | 5.96e-05 | 2.92e-04 | 2.24e-04 | 8.50e-08 | 0.501    | -1.41e-16 |
| ridge      | 0.544    | 0.612   | -0.067 | 0.024 [0.022, 0.026] | yes | 2.46e-04 | 1.94e-04 | 6.04e-08 | 0.428    | 5.97e-05 | 2.92e-04 | 2.24e-04 | 8.50e-08 | 0.501    | 1.67e-17  |
| lasso      | 0.542    | 0.603   | -0.062 | 0.016 [0.014, 0.017] | yes | 2.48e-04 | 1.96e-04 | 6.17e-08 | 0.408    | 6.26e-05 | 2.93e-04 | 2.24e-04 | 8.56e-08 | 0.474    | -5.89e-21 |
| elasticnet | 0.534    | 0.592   | -0.058 | 0.004 [0.003, 0.006] | yes | 2.52e-04 | 1.98e-04 | 6.34e-08 | 0.406    | 6.59e-05 | 2.95e-04 | 2.25e-04 | 8.70e-08 | 0.468    | 3.31e-21  |
| huber      | 0.543    | 0.620   | -0.077 | 0.033 [0.031, 0.034] | yes | 2.43e-04 | 1.91e-04 | 5.90e-08 | 0.451    | 4.62e-05 | 2.92e-04 | 2.23e-04 | 8.53e-08 | 0.531    | -1.46e-05 |
| poly2      | 0.561    | 0.620   | -0.059 | 0.033 [0.029, 0.036] | yes | 2.43e-04 | 1.92e-04 | 5.90e-08 | 0.347    | 7.21e-05 | 2.86e-04 | 2.19e-04 | 8.20e-08 | 0.410    | 8.59e-15  |
| poly3      | 0.562    | 0.615   | -0.054 | 0.028 [0.025, 0.031] | yes | 2.45e-04 | 1.93e-04 | 5.98e-08 | 0.431    | 7.46e-05 | 2.86e-04 | 2.19e-04 | 8.19e-08 | 0.418    | 2.41e-13  |

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| knn         | 0.586    | 0.594   | -0.008 | 0.007 [-0.000, 0.013]   | no  | 2.51e-04 | 1.99e-04 | 6.31e-08 | 0.352    | 8.43e-05 | 2.78e-04 | 2.12e-04 | 7.72e-08 | 0.376    | -2.41e-06 |
| svr ★       | 0.583    | 0.628   | -0.045 | 0.040 [0.033, 0.047]    | yes | 2.41e-04 | 1.89e-04 | 5.78e-08 | 0.356    | 6.62e-05 | 2.79e-04 | 2.12e-04 | 7.78e-08 | 0.411    | -1.41e-05 |
| kernelridge | 0.581    | 0.610   | -0.028 | 0.022 [0.011, 0.032]    | yes | 2.46e-04 | 1.94e-04 | 6.07e-08 | 0.500    | 7.68e-05 | 2.80e-04 | 2.12e-04 | 7.81e-08 | 0.696    | 1.60e-07  |
| gbm         | 0.477    | 0.482   | -0.005 | -0.106 [-0.114, -0.096] | no  | 2.84e-04 | 2.25e-04 | 8.06e-08 | 0.371    | 1.19e-04 | 3.12e-04 | 2.38e-04 | 9.76e-08 | 0.359    | -2.74e-07 |
| rf          | 0.596    | 0.608   | -0.012 | 0.020 [0.014, 0.026]    | yes | 2.47e-04 | 1.96e-04 | 6.09e-08 | 0.346    | 8.60e-05 | 2.75e-04 | 2.10e-04 | 7.54e-08 | 0.363    | 2.62e-08  |
| extratrees  | 0.615    | 0.608   | 0.007  | 0.020 [0.015, 0.025]    | yes | 2.47e-04 | 1.96e-04 | 6.10e-08 | 0.342    | 8.69e-05 | 2.68e-04 | 2.04e-04 | 7.18e-08 | 0.337    | 2.21e-19  |
| mlp         | 0.593    | 0.614   | -0.021 | 0.026 [0.018, 0.032]    | yes | 2.45e-04 | 1.95e-04 | 6.01e-08 | 0.357    | 8.42e-05 | 2.76e-04 | 2.11e-04 | 7.60e-08 | 0.371    | 2.63e-06  |
| gp          | 0.572    | 0.605   | -0.034 | 0.017 [0.007, 0.027]    | yes | 2.48e-04 | 1.95e-04 | 6.14e-08 | 0.346    | 8.20e-05 | 2.83e-04 | 2.15e-04 | 8.00e-08 | 0.394    | -2.86e-06 |

best-by-test-R2 = **svr** (0.628) · linear 0.612 · persistence 0.588 · H=60s · 1771.6s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=60s | horizon=60s | metric=AI

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.542    | 0.608   | -0.066 | 0.021 [0.018, 0.023]    | yes | 2.47e-04 | 1.95e-04 | 6.09e-08 | 0.435    | 5.84e-05 | 2.92e-04 | 2.24e-04 | 8.55e-08 | 0.506    | 6.29e-18  |
| ridge       | 0.542    | 0.608   | -0.066 | 0.021 [0.018, 0.023]    | yes | 2.47e-04 | 1.95e-04 | 6.09e-08 | 0.435    | 5.85e-05 | 2.92e-04 | 2.24e-04 | 8.55e-08 | 0.506    | 1.75e-17  |
| lasso       | 0.540    | 0.601   | -0.061 | 0.013 [0.011, 0.015]    | yes | 2.49e-04 | 1.96e-04 | 6.20e-08 | 0.411    | 6.31e-05 | 2.93e-04 | 2.24e-04 | 8.59e-08 | 0.477    | 6.63e-21  |
| elasticnet  | 0.534    | 0.591   | -0.056 | 0.003 [0.001, 0.005]    | yes | 2.52e-04 | 1.99e-04 | 6.36e-08 | 0.402    | 6.81e-05 | 2.95e-04 | 2.25e-04 | 8.69e-08 | 0.465    | 9.94e-21  |
| huber       | 0.541    | 0.616   | -0.076 | 0.029 [0.027, 0.030]    | yes | 2.44e-04 | 1.92e-04 | 5.96e-08 | 0.457    | 4.50e-05 | 2.93e-04 | 2.24e-04 | 8.58e-08 | 0.537    | -1.46e-05 |
| poly2       | 0.557    | 0.613   | -0.055 | 0.025 [0.021, 0.029]    | yes | 2.45e-04 | 1.94e-04 | 6.02e-08 | 0.350    | 7.34e-05 | 2.87e-04 | 2.20e-04 | 8.26e-08 | 0.406    | -9.05e-16 |
| poly3       | 0.561    | 0.609   | -0.048 | 0.021 [0.016, 0.025]    | yes | 2.47e-04 | 1.95e-04 | 6.08e-08 | 0.432    | 7.77e-05 | 2.86e-04 | 2.19e-04 | 8.20e-08 | 0.411    | 1.45e-14  |
| knn         | 0.583    | 0.594   | -0.011 | 0.006 [0.001, 0.012]    | yes | 2.51e-04 | 1.99e-04 | 6.31e-08 | 0.353    | 8.49e-05 | 2.79e-04 | 2.13e-04 | 7.79e-08 | 0.378    | -3.03e-06 |
| svr ★       | 0.580    | 0.625   | -0.045 | 0.038 [0.031, 0.044]    | yes | 2.41e-04 | 1.90e-04 | 5.83e-08 | 0.356    | 6.62e-05 | 2.80e-04 | 2.12e-04 | 7.84e-08 | 0.411    | -1.40e-05 |
| kernelridge | 0.578    | 0.607   | -0.029 | 0.020 [0.010, 0.028]    | yes | 2.47e-04 | 1.95e-04 | 6.10e-08 | 0.548    | 7.84e-05 | 2.81e-04 | 2.13e-04 | 7.87e-08 | 0.784    | 2.33e-07  |
| gbm         | 0.475    | 0.482   | -0.006 | -0.106 [-0.114, -0.096] | no  | 2.84e-04 | 2.25e-04 | 8.06e-08 | 0.371    | 1.18e-04 | 3.13e-04 | 2.39e-04 | 9.79e-08 | 0.361    | -2.74e-07 |
| rf          | 0.593    | 0.604   | -0.011 | 0.016 [0.009, 0.022]    | yes | 2.48e-04 | 1.97e-04 | 6.16e-08 | 0.345    | 8.68e-05 | 2.76e-04 | 2.11e-04 | 7.60e-08 | 0.364    | 3.44e-08  |
| extratrees  | 0.612    | 0.602   | 0.010  | 0.014 [0.008, 0.019]    | yes | 2.49e-04 | 1.98e-04 | 6.19e-08 | 0.343    | 8.93e-05 | 2.69e-04 | 2.05e-04 | 7.24e-08 | 0.338    | 2.11e-19  |
| mlp         | 0.589    | 0.615   | -0.025 | 0.027 [0.020, 0.032]    | yes | 2.45e-04 | 1.95e-04 | 5.99e-08 | 0.347    | 8.30e-05 | 2.77e-04 | 2.11e-04 | 7.66e-08 | 0.363    | 3.17e-06  |
| gp          | 0.569    | 0.602   | -0.033 | 0.015 [0.006, 0.023]    | yes | 2.49e-04 | 1.96e-04 | 6.18e-08 | 0.353    | 8.30e-05 | 2.84e-04 | 2.15e-04 | 8.04e-08 | 0.398    | -2.66e-06 |

best-by-test-R2 = **svr** (0.625) · linear 0.608 · persistence 0.588 · H=60s · 1814.1s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=2m Sigma=60s · horizon=2m (H=120s) — metrics: LE + AI

LE Sigma=60s | horizon=2m | metric=LE

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.580    | 0.656   | -0.076 | 0.025 [0.022, 0.028]    | yes | 2.99e-04 | 2.35e-04 | 8.95e-08 | 0.316    | 9.53e-05 | 3.57e-04 | 2.71e-04 | 1.27e-07 | 0.352    | -5.02e-16 |
| ridge       | 0.580    | 0.656   | -0.076 | 0.025 [0.022, 0.028]    | yes | 2.99e-04 | 2.35e-04 | 8.95e-08 | 0.316    | 9.54e-05 | 3.57e-04 | 2.71e-04 | 1.27e-07 | 0.352    | 1.34e-17  |
| lasso       | 0.578    | 0.649   | -0.071 | 0.018 [0.016, 0.021]    | yes | 3.02e-04 | 2.37e-04 | 9.12e-08 | 0.313    | 9.82e-05 | 3.57e-04 | 2.72e-04 | 1.28e-07 | 0.343    | -2.45e-20 |
| elasticnet  | 0.573    | 0.640   | -0.067 | 0.010 [0.008, 0.012]    | yes | 3.06e-04 | 2.40e-04 | 9.35e-08 | 0.314    | 1.02e-04 | 3.59e-04 | 2.73e-04 | 1.29e-07 | 0.341    | 1.21e-20  |
| huber       | 0.577    | 0.669   | -0.091 | 0.038 [0.036, 0.040]    | yes | 2.93e-04 | 2.30e-04 | 8.61e-08 | 0.315    | 7.76e-05 | 3.58e-04 | 2.70e-04 | 1.28e-07 | 0.359    | -2.19e-05 |
| poly2       | 0.597    | 0.661   | -0.064 | 0.030 [0.024, 0.035]    | yes | 2.97e-04 | 2.34e-04 | 8.82e-08 | 0.289    | 1.14e-04 | 3.49e-04 | 2.64e-04 | 1.22e-07 | 0.306    | -1.17e-15 |
| poly3       | 0.599    | 0.655   | -0.056 | 0.024 [0.019, 0.028]    | yes | 3.00e-04 | 2.36e-04 | 8.98e-08 | 0.325    | 1.18e-04 | 3.48e-04 | 2.64e-04 | 1.21e-07 | 0.305    | -9.43e-14 |
| knn         | 0.626    | 0.634   | -0.008 | 0.003 [-0.003, 0.009]   | no  | 3.08e-04 | 2.44e-04 | 9.51e-08 | 0.295    | 1.29e-04 | 3.36e-04 | 2.53e-04 | 1.13e-07 | 0.276    | -3.75e-06 |
| svr ★       | 0.622    | 0.676   | -0.054 | 0.045 [0.040, 0.051]    | yes | 2.90e-04 | 2.28e-04 | 8.43e-08 | 0.283    | 1.03e-04 | 3.38e-04 | 2.53e-04 | 1.14e-07 | 0.287    | -2.30e-05 |
| kernelridge | 0.621    | 0.651   | -0.030 | 0.021 [0.010, 0.030]    | yes | 3.01e-04 | 2.36e-04 | 9.06e-08 | 0.426    | 1.19e-04 | 3.38e-04 | 2.53e-04 | 1.15e-07 | 0.521    | -2.93e-06 |
| gbm         | 0.510    | 0.504   | 0.006  | -0.127 [-0.136, -0.116] | no  | 3.59e-04 | 2.86e-04 | 1.29e-07 | 0.358    | 1.77e-04 | 3.85e-04 | 2.91e-04 | 1.48e-07 | 0.308    | -1.45e-07 |
| rf          | 0.635    | 0.645   | -0.010 | 0.014 [0.006, 0.023]    | yes | 3.04e-04 | 2.42e-04 | 9.22e-08 | 0.295    | 1.31e-04 | 3.32e-04 | 2.50e-04 | 1.10e-07 | 0.270    | 1.19e-07  |
| extratrees  | 0.654    | 0.644   | 0.010  | 0.013 [0.005, 0.019]    | yes | 3.04e-04 | 2.42e-04 | 9.27e-08 | 0.294    | 1.33e-04 | 3.24e-04 | 2.43e-04 | 1.05e-07 | 0.252    | 2.52e-20  |
| mlp         | 0.633    | 0.653   | -0.020 | 0.022 [0.012, 0.031]    | yes | 3.00e-04 | 2.39e-04 | 9.03e-08 | 0.291    | 1.29e-04 | 3.33e-04 | 2.51e-04 | 1.11e-07 | 0.276    | -1.72e-06 |
| gp          | 0.616    | 0.649   | -0.033 | 0.018 [0.012, 0.024]    | yes | 3.02e-04 | 2.38e-04 | 9.12e-08 | 0.295    | 1.26e-04 | 3.41e-04 | 2.56e-04 | 1.16e-07 | 0.295    | -4.29e-06 |

best-by-test-R2 = **svr** (0.676) · linear 0.656 · persistence 0.631 · H=120s · 1780.9s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=60s | horizon=2m | metric=AI

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.577    | 0.652   | -0.075 | 0.021 [0.018, 0.025]    | yes | 3.01e-04 | 2.36e-04 | 9.04e-08 | 0.320    | 9.51e-05 | 3.58e-04 | 2.72e-04 | 1.28e-07 | 0.356    | -4.22e-17 |
| ridge       | 0.577    | 0.652   | -0.075 | 0.021 [0.018, 0.025]    | yes | 3.01e-04 | 2.36e-04 | 9.04e-08 | 0.320    | 9.52e-05 | 3.58e-04 | 2.72e-04 | 1.28e-07 | 0.355    | 1.34e-17  |
| lasso       | 0.576    | 0.646   | -0.070 | 0.015 [0.011, 0.018]    | yes | 3.03e-04 | 2.39e-04 | 9.21e-08 | 0.314    | 9.98e-05 | 3.58e-04 | 2.73e-04 | 1.28e-07 | 0.345    | 2.13e-21  |
| elasticnet  | 0.572    | 0.638   | -0.065 | 0.007 [0.004, 0.009]    | yes | 3.07e-04 | 2.41e-04 | 9.42e-08 | 0.315    | 1.05e-04 | 3.60e-04 | 2.73e-04 | 1.29e-07 | 0.341    | -4.98e-20 |
| huber       | 0.575    | 0.664   | -0.089 | 0.033 [0.031, 0.035]    | yes | 2.95e-04 | 2.31e-04 | 8.73e-08 | 0.319    | 7.70e-05 | 3.59e-04 | 2.71e-04 | 1.29e-07 | 0.364    | -2.19e-05 |
| poly2       | 0.594    | 0.651   | -0.057 | 0.020 [0.014, 0.026]    | yes | 3.01e-04 | 2.38e-04 | 9.07e-08 | 0.294    | 1.17e-04 | 3.51e-04 | 2.65e-04 | 1.23e-07 | 0.305    | -5.66e-15 |
| poly3       | 0.599    | 0.646   | -0.047 | 0.015 [0.009, 0.021]    | yes | 3.03e-04 | 2.40e-04 | 9.20e-08 | 0.330    | 1.24e-04 | 3.48e-04 | 2.64e-04 | 1.21e-07 | 0.302    | 2.03e-13  |
| knn         | 0.624    | 0.634   | -0.010 | 0.003 [-0.005, 0.009]   | no  | 3.09e-04 | 2.44e-04 | 9.52e-08 | 0.297    | 1.31e-04 | 3.37e-04 | 2.54e-04 | 1.14e-07 | 0.277    | -4.52e-06 |
| svr ★       | 0.619    | 0.671   | -0.052 | 0.040 [0.034, 0.045]    | yes | 2.93e-04 | 2.30e-04 | 8.56e-08 | 0.285    | 1.03e-04 | 3.40e-04 | 2.54e-04 | 1.15e-07 | 0.288    | -2.29e-05 |
| kernelridge | 0.618    | 0.647   | -0.029 | 0.017 [0.008, 0.025]    | yes | 3.03e-04 | 2.38e-04 | 9.16e-08 | 0.461    | 1.21e-04 | 3.40e-04 | 2.54e-04 | 1.15e-07 | 0.533    | -3.06e-06 |
| gbm         | 0.508    | 0.504   | 0.004  | -0.127 [-0.136, -0.116] | no  | 3.59e-04 | 2.86e-04 | 1.29e-07 | 0.358    | 1.76e-04 | 3.86e-04 | 2.91e-04 | 1.49e-07 | 0.309    | -1.41e-07 |
| rf          | 0.632    | 0.640   | -0.009 | 0.010 [-0.000, 0.018]   | no  | 3.06e-04 | 2.43e-04 | 9.34e-08 | 0.298    | 1.34e-04 | 3.34e-04 | 2.51e-04 | 1.11e-07 | 0.271    | 1.27e-07  |
| extratrees  | 0.651    | 0.636   | 0.015  | 0.006 [-0.004, 0.013]   | no  | 3.07e-04 | 2.45e-04 | 9.45e-08 | 0.298    | 1.38e-04 | 3.25e-04 | 2.44e-04 | 1.06e-07 | 0.252    | 4.17e-20  |
| mlp         | 0.630    | 0.654   | -0.025 | 0.024 [0.014, 0.032]    | yes | 3.00e-04 | 2.38e-04 | 8.98e-08 | 0.289    | 1.26e-04 | 3.35e-04 | 2.52e-04 | 1.12e-07 | 0.274    | -2.74e-06 |
| gp          | 0.613    | 0.641   | -0.028 | 0.011 [0.003, 0.017]    | yes | 3.05e-04 | 2.41e-04 | 9.32e-08 | 0.297    | 1.28e-04 | 3.42e-04 | 2.57e-04 | 1.17e-07 | 0.297    | -4.50e-06 |

best-by-test-R2 = **svr** (0.671) · linear 0.652 · persistence 0.631 · H=120s · 1836.1s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=5m Sigma=60s · horizon=5m (H=300s) — metrics: LE + AI

LE Sigma=60s | horizon=5m | metric=LE

| learner | train R2 | test R2 | gap    | dR2 [CI95]           | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|---------|----------|---------|--------|----------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear  | 0.626    | 0.703   | -0.077 | 0.027 [0.023, 0.031] | yes | 4.12e-04 | 3.26e-04 | 1.70e-07 | 0.225    | 1.67e-04 | 4.89e-04 | 3.69e-04 | 2.39e-07 | 0.246    | -1.96e-16 |

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| ridge       | 0.626    | 0.703   | -0.077 | 0.027 [0.023, 0.031]    | yes | 4.12e-04 | 3.26e-04 | 1.70e-07 | 0.225    | 1.67e-04 | 4.89e-04 | 3.69e-04 | 2.39e-07 | 0.246    | 1.98e-17  |
| lasso       | 0.625    | 0.698   | -0.073 | 0.022 [0.019, 0.026]    | yes | 4.15e-04 | 3.28e-04 | 1.73e-07 | 0.225    | 1.70e-04 | 4.89e-04 | 3.69e-04 | 2.39e-07 | 0.243    | -4.41e-20 |
| elasticnet  | 0.623    | 0.692   | -0.069 | 0.016 [0.013, 0.019]    | yes | 4.20e-04 | 3.31e-04 | 1.76e-07 | 0.226    | 1.73e-04 | 4.91e-04 | 3.71e-04 | 2.41e-07 | 0.242    | -8.53e-21 |
| huber       | 0.623    | 0.720   | -0.097 | 0.044 [0.042, 0.046]    | yes | 4.00e-04 | 3.13e-04 | 1.60e-07 | 0.218    | 1.36e-04 | 4.91e-04 | 3.66e-04 | 2.41e-07 | 0.251    | -3.85e-05 |
| poly2       | 0.647    | 0.705   | -0.058 | 0.029 [0.022, 0.036]    | yes | 4.11e-04 | 3.27e-04 | 1.69e-07 | 0.218    | 1.97e-04 | 4.75e-04 | 3.56e-04 | 2.26e-07 | 0.216    | 2.17e-15  |
| poly3       | 0.649    | 0.697   | -0.048 | 0.021 [0.016, 0.027]    | yes | 4.16e-04 | 3.31e-04 | 1.73e-07 | 0.240    | 2.04e-04 | 4.73e-04 | 3.55e-04 | 2.24e-07 | 0.213    | 3.79e-13  |
| knn         | 0.682    | 0.674   | 0.008  | -0.002 [-0.009, 0.006]  | no  | 4.32e-04 | 3.45e-04 | 1.86e-07 | 0.224    | 2.21e-04 | 4.51e-04 | 3.33e-04 | 2.03e-07 | 0.187    | -4.33e-06 |
| svr ★       | 0.677    | 0.733   | -0.056 | 0.058 [0.052, 0.065]    | yes | 3.90e-04 | 3.07e-04 | 1.52e-07 | 0.200    | 1.57e-04 | 4.54e-04 | 3.33e-04 | 2.06e-07 | 0.197    | -4.02e-05 |
| kernelridge | 0.678    | 0.699   | -0.022 | 0.024 [0.014, 0.033]    | yes | 4.15e-04 | 3.29e-04 | 1.72e-07 | 0.281    | 2.02e-04 | 4.54e-04 | 3.35e-04 | 2.06e-07 | 0.283    | -2.30e-06 |
| gbm         | 0.601    | 0.597   | 0.004  | -0.079 [-0.087, -0.070] | no  | 4.80e-04 | 3.88e-04 | 2.30e-07 | 0.272    | 2.62e-04 | 5.05e-04 | 3.79e-04 | 2.55e-07 | 0.219    | 1.72e-08  |
| rf          | 0.708    | 0.675   | 0.033  | -0.001 [-0.014, 0.011]  | no  | 4.31e-04 | 3.48e-04 | 1.86e-07 | 0.231    | 2.31e-04 | 4.32e-04 | 3.20e-04 | 1.87e-07 | 0.175    | 5.53e-08  |
| extratrees  | 0.708    | 0.682   | 0.027  | 0.006 [-0.005, 0.015]   | no  | 4.27e-04 | 3.45e-04 | 1.82e-07 | 0.227    | 2.28e-04 | 4.32e-04 | 3.19e-04 | 1.86e-07 | 0.170    | -3.96e-20 |
| mlp         | 0.689    | 0.696   | -0.006 | 0.020 [0.007, 0.031]    | yes | 4.17e-04 | 3.36e-04 | 1.74e-07 | 0.216    | 2.17e-04 | 4.45e-04 | 3.30e-04 | 1.98e-07 | 0.184    | 2.29e-06  |
| gp          | 0.665    | 0.695   | -0.029 | 0.019 [0.010, 0.028]    | yes | 4.18e-04 | 3.32e-04 | 1.75e-07 | 0.217    | 2.04e-04 | 4.62e-04 | 3.40e-04 | 2.14e-07 | 0.195    | -2.92e-06 |

best-by-test-R2 = **svr** (0.733) · linear 0.703 · persistence 0.676 · H=300s · 1803.2s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=60s | horizon=5m | metric=AI

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.623    | 0.698   | -0.075 | 0.023 [0.018, 0.027]    | yes | 4.15e-04 | 3.28e-04 | 1.72e-07 | 0.227    | 1.68e-04 | 4.90e-04 | 3.71e-04 | 2.41e-07 | 0.249    | -5.21e-16 |
| ridge       | 0.623    | 0.698   | -0.075 | 0.023 [0.018, 0.027]    | yes | 4.15e-04 | 3.28e-04 | 1.72e-07 | 0.227    | 1.68e-04 | 4.90e-04 | 3.71e-04 | 2.41e-07 | 0.249    | 1.96e-17  |
| lasso       | 0.623    | 0.694   | -0.071 | 0.018 [0.013, 0.022]    | yes | 4.18e-04 | 3.30e-04 | 1.75e-07 | 0.227    | 1.73e-04 | 4.91e-04 | 3.71e-04 | 2.41e-07 | 0.245    | -5.12e-20 |
| elasticnet  | 0.621    | 0.688   | -0.067 | 0.012 [0.008, 0.016]    | yes | 4.22e-04 | 3.33e-04 | 1.78e-07 | 0.228    | 1.78e-04 | 4.92e-04 | 3.72e-04 | 2.42e-07 | 0.243    | -2.84e-20 |
| huber       | 0.620    | 0.715   | -0.094 | 0.039 [0.036, 0.041]    | yes | 4.04e-04 | 3.16e-04 | 1.63e-07 | 0.222    | 1.36e-04 | 4.92e-04 | 3.68e-04 | 2.42e-07 | 0.255    | -3.86e-05 |
| poly2       | 0.643    | 0.693   | -0.050 | 0.017 [0.010, 0.024]    | yes | 4.19e-04 | 3.33e-04 | 1.76e-07 | 0.224    | 2.03e-04 | 4.77e-04 | 3.57e-04 | 2.28e-07 | 0.218    | 5.35e-15  |
| poly3       | 0.650    | 0.687   | -0.037 | 0.011 [0.003, 0.019]    | yes | 4.23e-04 | 3.38e-04 | 1.79e-07 | 0.245    | 2.14e-04 | 4.73e-04 | 3.54e-04 | 2.24e-07 | 0.222    | 6.07e-13  |
| knn         | 0.679    | 0.671   | 0.008  | -0.005 [-0.015, 0.003]  | no  | 4.34e-04 | 3.48e-04 | 1.88e-07 | 0.228    | 2.26e-04 | 4.53e-04 | 3.35e-04 | 2.05e-07 | 0.189    | -5.68e-06 |
| svr ★       | 0.674    | 0.725   | -0.051 | 0.049 [0.042, 0.057]    | yes | 3.96e-04 | 3.12e-04 | 1.57e-07 | 0.203    | 1.67e-04 | 4.56e-04 | 3.34e-04 | 2.08e-07 | 0.198    | -3.97e-05 |
| kernelridge | 0.675    | 0.695   | -0.019 | 0.019 [0.008, 0.028]    | yes | 4.18e-04 | 3.34e-04 | 1.75e-07 | 0.307    | 2.09e-04 | 4.55e-04 | 3.36e-04 | 2.07e-07 | 0.299    | -1.95e-06 |
| gbm         | 0.598    | 0.591   | 0.007  | -0.085 [-0.092, -0.075] | no  | 4.84e-04 | 3.91e-04 | 2.34e-07 | 0.274    | 2.65e-04 | 5.07e-04 | 3.81e-04 | 2.57e-07 | 0.221    | 3.20e-08  |
| rf          | 0.715    | 0.658   | 0.056  | -0.017 [-0.032, -0.004] | no  | 4.42e-04 | 3.57e-04 | 1.95e-07 | 0.238    | 2.43e-04 | 4.27e-04 | 3.16e-04 | 1.82e-07 | 0.170    | 6.66e-08  |
| extratrees  | 0.705    | 0.670   | 0.035  | -0.006 [-0.019, 0.005]  | no  | 4.34e-04 | 3.52e-04 | 1.89e-07 | 0.233    | 2.38e-04 | 4.34e-04 | 3.21e-04 | 1.88e-07 | 0.171    | 1.51e-20  |
| mlp         | 0.685    | 0.691   | -0.006 | 0.015 [0.002, 0.027]    | yes | 4.20e-04 | 3.38e-04 | 1.77e-07 | 0.220    | 2.20e-04 | 4.49e-04 | 3.32e-04 | 2.01e-07 | 0.187    | -5.20e-06 |
| gp          | 0.664    | 0.689   | -0.024 | 0.013 [0.003, 0.021]    | yes | 4.22e-04 | 3.37e-04 | 1.78e-07 | 0.220    | 2.13e-04 | 4.63e-04 | 3.41e-04 | 2.14e-07 | 0.196    | -1.72e-06 |

best-by-test-R2 = **svr** (0.725) · linear 0.698 · persistence 0.676 · H=300s · 1850.7s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=10m Sigma=60s · horizon=10m (H=600s) — metrics: LE + AI

LE Sigma=60s | horizon=10m | metric=LE

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.632    | 0.711   | -0.079 | 0.029 [0.024, 0.033]    | yes | 5.61e-04 | 4.45e-04 | 3.15e-07 | 0.191    | 2.54e-04 | 6.60e-04 | 4.96e-04 | 4.36e-07 | 0.210    | 5.56e-16  |
| ridge       | 0.632    | 0.711   | -0.079 | 0.029 [0.024, 0.033]    | yes | 5.61e-04 | 4.46e-04 | 3.15e-07 | 0.191    | 2.54e-04 | 6.60e-04 | 4.96e-04 | 4.36e-07 | 0.210    | -4.74e-17 |
| lasso       | 0.631    | 0.708   | -0.076 | 0.025 [0.021, 0.029]    | yes | 5.64e-04 | 4.48e-04 | 3.18e-07 | 0.192    | 2.57e-04 | 6.61e-04 | 4.97e-04 | 4.37e-07 | 0.208    | 1.13e-19  |
| elasticnet  | 0.630    | 0.703   | -0.073 | 0.021 [0.017, 0.025]    | yes | 5.68e-04 | 4.51e-04 | 3.23e-07 | 0.193    | 2.60e-04 | 6.62e-04 | 4.98e-04 | 4.38e-07 | 0.208    | 1.73e-20  |
| huber       | 0.628    | 0.733   | -0.105 | 0.051 [0.048, 0.053]    | yes | 5.39e-04 | 4.23e-04 | 2.91e-07 | 0.182    | 2.05e-04 | 6.64e-04 | 4.92e-04 | 4.41e-07 | 0.216    | -6.07e-05 |
| poly2       | 0.654    | 0.710   | -0.056 | 0.028 [0.020, 0.035]    | yes | 5.62e-04 | 4.50e-04 | 3.16e-07 | 0.190    | 2.95e-04 | 6.40e-04 | 4.76e-04 | 4.09e-07 | 0.187    | 4.13e-15  |
| poly3       | 0.657    | 0.701   | -0.043 | 0.018 [0.012, 0.025]    | yes | 5.71e-04 | 4.58e-04 | 3.26e-07 | 0.218    | 3.07e-04 | 6.37e-04 | 4.74e-04 | 4.06e-07 | 0.182    | 1.23e-12  |
| knn         | 0.694    | 0.678   | 0.016  | -0.004 [-0.013, 0.004]  | no  | 5.92e-04 | 4.79e-04 | 3.50e-07 | 0.197    | 3.30e-04 | 6.02e-04 | 4.41e-04 | 3.63e-07 | 0.158    | -6.37e-06 |
| svr ★       | 0.687    | 0.748   | -0.061 | 0.066 [0.059, 0.074]    | yes | 5.23e-04 | 4.13e-04 | 2.74e-07 | 0.167    | 2.28e-04 | 6.09e-04 | 4.40e-04 | 3.71e-07 | 0.167    | -6.26e-05 |
| kernelridge | 0.687    | 0.703   | -0.016 | 0.021 [0.008, 0.033]    | yes | 5.68e-04 | 4.53e-04 | 3.23e-07 | 0.356    | 2.98e-04 | 6.09e-04 | 4.44e-04 | 3.71e-07 | 0.283    | -2.28e-06 |
| gbm         | 0.632    | 0.625   | 0.008  | -0.057 [-0.065, -0.049] | no  | 6.39e-04 | 5.22e-04 | 4.08e-07 | 0.232    | 3.69e-04 | 6.60e-04 | 4.92e-04 | 4.36e-07 | 0.181    | 2.17e-07  |
| rf          | 0.730    | 0.661   | 0.069  | -0.022 [-0.039, -0.007] | no  | 6.08e-04 | 4.97e-04 | 3.69e-07 | 0.210    | 3.59e-04 | 5.66e-04 | 4.14e-04 | 3.20e-07 | 0.140    | 2.06e-07  |
| extratrees  | 0.720    | 0.678   | 0.042  | -0.004 [-0.018, 0.006]  | no  | 5.92e-04 | 4.84e-04 | 3.51e-07 | 0.203    | 3.45e-04 | 5.76e-04 | 4.21e-04 | 3.32e-07 | 0.142    | -2.93e-20 |
| mlp         | 0.701    | 0.694   | 0.007  | 0.012 [-0.004, 0.025]   | no  | 5.77e-04 | 4.70e-04 | 3.33e-07 | 0.191    | 3.28e-04 | 5.95e-04 | 4.37e-04 | 3.55e-07 | 0.154    | 7.58e-06  |
| gp          | 0.674    | 0.692   | -0.018 | 0.010 [-0.001, 0.020]   | no  | 5.79e-04 | 4.66e-04 | 3.35e-07 | 0.191    | 3.17e-04 | 6.21e-04 | 4.52e-04 | 3.86e-07 | 0.165    | -3.72e-06 |

best-by-test-R2 = **svr** (0.748) · linear 0.711 · persistence 0.682 · H=600s · 1803.3s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI Sigma=60s | horizon=10m | metric=AI

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.629    | 0.706   | -0.077 | 0.024 [0.019, 0.029]    | yes | 5.65e-04 | 4.49e-04 | 3.20e-07 | 0.194    | 2.57e-04 | 6.63e-04 | 4.99e-04 | 4.40e-07 | 0.213    | -1.39e-15 |
| ridge       | 0.629    | 0.706   | -0.077 | 0.024 [0.018, 0.029]    | yes | 5.65e-04 | 4.49e-04 | 3.20e-07 | 0.194    | 2.57e-04 | 6.63e-04 | 4.99e-04 | 4.40e-07 | 0.213    | -4.85e-17 |
| lasso       | 0.628    | 0.703   | -0.074 | 0.021 [0.015, 0.025]    | yes | 5.69e-04 | 4.51e-04 | 3.23e-07 | 0.194    | 2.61e-04 | 6.63e-04 | 4.99e-04 | 4.40e-07 | 0.211    | 1.95e-19  |
| elasticnet  | 0.627    | 0.698   | -0.071 | 0.016 [0.011, 0.021]    | yes | 5.73e-04 | 4.55e-04 | 3.28e-07 | 0.195    | 2.66e-04 | 6.64e-04 | 5.00e-04 | 4.41e-07 | 0.209    | 1.19e-19  |
| huber       | 0.625    | 0.727   | -0.102 | 0.045 [0.042, 0.048]    | yes | 5.45e-04 | 4.27e-04 | 2.97e-07 | 0.186    | 2.06e-04 | 6.67e-04 | 4.94e-04 | 4.44e-07 | 0.219    | -6.09e-05 |
| poly2       | 0.651    | 0.697   | -0.046 | 0.015 [0.006, 0.023]    | yes | 5.74e-04 | 4.60e-04 | 3.30e-07 | 0.195    | 3.05e-04 | 6.43e-04 | 4.79e-04 | 4.14e-07 | 0.188    | 2.05e-14  |
| poly3       | 0.658    | 0.690   | -0.032 | 0.008 [-0.002, 0.017]   | no  | 5.80e-04 | 4.69e-04 | 3.37e-07 | 0.223    | 3.22e-04 | 6.36e-04 | 4.72e-04 | 4.05e-07 | 0.190    | 5.14e-13  |
| knn         | 0.691    | 0.672   | 0.019  | -0.011 [-0.022, -0.002] | no  | 5.98e-04 | 4.85e-04 | 3.57e-07 | 0.201    | 3.39e-04 | 6.05e-04 | 4.43e-04 | 3.67e-07 | 0.159    | -8.04e-06 |
| svr ★       | 0.685    | 0.736   | -0.051 | 0.054 [0.045, 0.061]    | yes | 5.36e-04 | 4.26e-04 | 2.87e-07 | 0.173    | 2.49e-04 | 6.11e-04 | 4.41e-04 | 3.74e-07 | 0.168    | -6.22e-05 |
| kernelridge | 0.685    | 0.696   | -0.011 | 0.014 [0.002, 0.024]    | yes | 5.75e-04 | 4.61e-04 | 3.31e-07 | 0.340    | 3.08e-04 | 6.11e-04 | 4.46e-04 | 3.73e-07 | 0.274    | -1.86e-06 |
| gbm         | 0.629    | 0.618   | 0.011  | -0.064 [-0.071, -0.055] | no  | 6.44e-04 | 5.27e-04 | 4.15e-07 | 0.235    | 3.76e-04 | 6.63e-04 | 4.94e-04 | 4.39e-07 | 0.183    | 2.25e-07  |
| rf          | 0.726    | 0.648   | 0.078  | -0.034 [-0.052, -0.017] | no  | 6.18e-04 | 5.06e-04 | 3.82e-07 | 0.215    | 3.70e-04 | 5.70e-04 | 4.17e-04 | 3.24e-07 | 0.141    | 1.99e-07  |
| extratrees  | 0.717    | 0.664   | 0.054  | -0.019 [-0.034, -0.005] | no  | 6.05e-04 | 4.96e-04 | 3.66e-07 | 0.210    | 3.61e-04 | 5.79e-04 | 4.23e-04 | 3.35e-07 | 0.143    | -3.50e-20 |
| mlp         | 0.697    | 0.688   | 0.009  | 0.006 [-0.010, 0.020]   | no  | 5.83e-04 | 4.76e-04 | 3.40e-07 | 0.195    | 3.39e-04 | 5.99e-04 | 4.40e-04 | 3.59e-07 | 0.155    | 7.65e-06  |
| gp          | 0.677    | 0.686   | -0.009 | 0.004 [-0.008, 0.014]   | no  | 5.85e-04 | 4.72e-04 | 3.42e-07 | 0.196    | 3.24e-04 | 6.19e-04 | 4.52e-04 | 3.83e-07 | 0.164    | -3.63e-06 |

LE

Sigma=60s | horizon=15m | metric=LE

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.622    | 0.707   | -0.085 | 0.031 [0.026, 0.035]    | yes | 6.85e-04 | 5.46e-04 | 4.69e-07 | 0.182    | 3.27e-04 | 8.05e-04 | 6.02e-04 | 6.48e-07 | 0.200    | -7.75e-17 |
| ridge       | 0.622    | 0.707   | -0.085 | 0.031 [0.026, 0.035]    | yes | 6.85e-04 | 5.46e-04 | 4.69e-07 | 0.182    | 3.27e-04 | 8.05e-04 | 6.02e-04 | 6.48e-07 | 0.200    | 5.23e-17  |
| lasso       | 0.622    | 0.704   | -0.082 | 0.028 [0.023, 0.032]    | yes | 6.88e-04 | 5.48e-04 | 4.73e-07 | 0.183    | 3.30e-04 | 8.05e-04 | 6.02e-04 | 6.49e-07 | 0.199    | 2.95e-19  |
| elasticnet  | 0.621    | 0.700   | -0.080 | 0.024 [0.020, 0.028]    | yes | 6.92e-04 | 5.51e-04 | 4.79e-07 | 0.184    | 3.33e-04 | 8.06e-04 | 6.03e-04 | 6.50e-07 | 0.198    | 2.99e-19  |
| huber       | 0.617    | 0.732   | -0.115 | 0.056 [0.054, 0.059]    | yes | 6.54e-04 | 5.14e-04 | 4.28e-07 | 0.172    | 2.62e-04 | 8.10e-04 | 5.95e-04 | 6.57e-07 | 0.206    | -7.99e-05 |
| poly2       | 0.645    | 0.703   | -0.058 | 0.027 [0.018, 0.035]    | yes | 6.89e-04 | 5.55e-04 | 4.75e-07 | 0.183    | 3.77e-04 | 7.80e-04 | 5.77e-04 | 6.09e-07 | 0.180    | 2.38e-15  |
| poly3       | 0.648    | 0.692   | -0.044 | 0.016 [0.009, 0.022]    | yes | 7.01e-04 | 5.66e-04 | 4.92e-07 | 0.212    | 3.92e-04 | 7.77e-04 | 5.74e-04 | 6.03e-07 | 0.174    | 8.48e-14  |
| knn         | 0.687    | 0.667   | 0.020  | -0.009 [-0.019, -0.000] | no  | 7.30e-04 | 5.93e-04 | 5.32e-07 | 0.191    | 4.20e-04 | 7.33e-04 | 5.32e-04 | 5.37e-07 | 0.151    | -6.73e-06 |
| svr ★       | 0.680    | 0.737   | -0.058 | 0.061 [0.052, 0.070]    | yes | 6.48e-04 | 5.16e-04 | 4.19e-07 | 0.162    | 3.07e-04 | 7.41e-04 | 5.29e-04 | 5.49e-07 | 0.160    | -7.79e-05 |
| kernelridge | 0.681    | 0.698   | -0.017 | 0.022 [0.009, 0.031]    | yes | 6.95e-04 | 5.63e-04 | 4.83e-07 | 0.222    | 3.92e-04 | 7.40e-04 | 5.38e-04 | 5.48e-07 | 0.206    | 6.39e-06  |
| gbm         | 0.635    | 0.624   | 0.011  | -0.052 [-0.060, -0.043] | no  | 7.75e-04 | 6.38e-04 | 6.01e-07 | 0.222    | 4.64e-04 | 7.91e-04 | 5.85e-04 | 6.26e-07 | 0.171    | 2.54e-07  |
| rf          | 0.712    | 0.649   | 0.063  | -0.027 [-0.046, -0.012] | no  | 7.49e-04 | 6.14e-04 | 5.61e-07 | 0.205    | 4.51e-04 | 7.03e-04 | 5.12e-04 | 4.95e-07 | 0.141    | 1.33e-07  |
| extratrees  | 0.714    | 0.663   | 0.051  | -0.013 [-0.029, -0.001] | no  | 7.34e-04 | 6.05e-04 | 5.39e-07 | 0.200    | 4.44e-04 | 7.01e-04 | 5.08e-04 | 4.91e-07 | 0.135    | -3.56e-20 |
| mlp         | 0.692    | 0.678   | 0.014  | 0.002 [-0.015, 0.016]   | no  | 7.17e-04 | 5.90e-04 | 5.15e-07 | 0.189    | 4.33e-04 | 7.27e-04 | 5.30e-04 | 5.29e-07 | 0.148    | 1.33e-05  |
| gp          | 0.671    | 0.680   | -0.009 | 0.004 [-0.011, 0.016]   | no  | 7.15e-04 | 5.72e-04 | 5.12e-07 | 0.182    | 3.92e-04 | 7.52e-04 | 5.46e-04 | 5.65e-07 | 0.156    | 8.27e-06  |

best-by-test-R2 = **svr** (0.737) · linear 0.707 · persistence 0.676 · H=900s · 1777.1s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI

Sigma=60s | horizon=15m | metric=AI

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.619    | 0.702   | -0.083 | 0.026 [0.019, 0.031]    | yes | 6.90e-04 | 5.50e-04 | 4.76e-07 | 0.185    | 3.31e-04 | 8.09e-04 | 6.05e-04 | 6.54e-07 | 0.203    | -2.34e-17 |
| ridge       | 0.619    | 0.702   | -0.083 | 0.026 [0.019, 0.031]    | yes | 6.90e-04 | 5.50e-04 | 4.76e-07 | 0.185    | 3.31e-04 | 8.09e-04 | 6.05e-04 | 6.54e-07 | 0.203    | 5.21e-17  |
| lasso       | 0.619    | 0.699   | -0.080 | 0.023 [0.017, 0.028]    | yes | 6.94e-04 | 5.53e-04 | 4.81e-07 | 0.185    | 3.36e-04 | 8.09e-04 | 6.05e-04 | 6.54e-07 | 0.201    | 2.62e-19  |
| elasticnet  | 0.618    | 0.695   | -0.077 | 0.019 [0.013, 0.024]    | yes | 6.98e-04 | 5.56e-04 | 4.87e-07 | 0.187    | 3.41e-04 | 8.10e-04 | 6.06e-04 | 6.55e-07 | 0.200    | 2.22e-19  |
| huber       | 0.614    | 0.726   | -0.112 | 0.050 [0.047, 0.053]    | yes | 6.61e-04 | 5.19e-04 | 4.37e-07 | 0.176    | 2.63e-04 | 8.14e-04 | 5.99e-04 | 6.62e-07 | 0.210    | -8.00e-05 |
| poly2       | 0.641    | 0.689   | -0.048 | 0.013 [0.004, 0.022]    | yes | 7.05e-04 | 5.67e-04 | 4.96e-07 | 0.189    | 3.88e-04 | 7.85e-04 | 5.81e-04 | 6.16e-07 | 0.182    | 1.51e-14  |
| poly3       | 0.650    | 0.682   | -0.032 | 0.006 [-0.005, 0.016]   | no  | 7.13e-04 | 5.79e-04 | 5.08e-07 | 0.217    | 4.10e-04 | 7.75e-04 | 5.71e-04 | 6.01e-07 | 0.194    | 4.54e-13  |
| knn         | 0.684    | 0.659   | 0.025  | -0.017 [-0.030, -0.007] | no  | 7.38e-04 | 6.02e-04 | 5.44e-07 | 0.196    | 4.33e-04 | 7.36e-04 | 5.35e-04 | 5.42e-07 | 0.152    | -8.69e-06 |
| svr ★       | 0.677    | 0.732   | -0.055 | 0.056 [0.046, 0.065]    | yes | 6.54e-04 | 5.22e-04 | 4.28e-07 | 0.165    | 3.16e-04 | 7.44e-04 | 5.32e-04 | 5.54e-07 | 0.161    | -7.55e-05 |
| kernelridge | 0.678    | 0.686   | -0.007 | 0.009 [-0.005, 0.020]   | no  | 7.09e-04 | 5.75e-04 | 5.03e-07 | 0.231    | 4.03e-04 | 7.43e-04 | 5.40e-04 | 5.52e-07 | 0.235    | 6.97e-06  |
| gbm         | 0.631    | 0.621   | 0.010  | -0.055 [-0.064, -0.047] | no  | 7.78e-04 | 6.40e-04 | 6.06e-07 | 0.223    | 4.65e-04 | 7.95e-04 | 5.89e-04 | 6.33e-07 | 0.173    | 2.20e-07  |
| rf          | 0.720    | 0.628   | 0.092  | -0.048 [-0.068, -0.029] | no  | 7.71e-04 | 6.34e-04 | 5.94e-07 | 0.213    | 4.79e-04 | 6.93e-04 | 5.03e-04 | 4.80e-07 | 0.134    | 1.67e-07  |
| extratrees  | 0.711    | 0.648   | 0.063  | -0.028 [-0.045, -0.013] | no  | 7.50e-04 | 6.19e-04 | 5.62e-07 | 0.206    | 4.64e-04 | 7.04e-04 | 5.10e-04 | 4.95e-07 | 0.136    | 3.63e-22  |
| mlp         | 0.690    | 0.667   | 0.023  | -0.009 [-0.028, 0.008]  | no  | 7.29e-04 | 6.00e-04 | 5.32e-07 | 0.194    | 4.45e-04 | 7.29e-04 | 5.30e-04 | 5.31e-07 | 0.150    | 3.41e-06  |
| gp          | 0.670    | 0.684   | -0.014 | 0.008 [-0.005, 0.018]   | no  | 7.11e-04 | 5.77e-04 | 5.05e-07 | 0.186    | 4.10e-04 | 7.53e-04 | 5.47e-04 | 5.67e-07 | 0.157    | 9.42e-06  |

best-by-test-R2 = **svr** (0.732) · linear 0.702 · persistence 0.676 · H=900s · 1859.8s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

LE

Sigma=60s | horizon=30m | metric=LE

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.582    | 0.681   | -0.100 | 0.036 [0.030, 0.041]    | yes | 9.93e-04 | 7.99e-04 | 9.86e-07 | 0.180    | 5.17e-04 | 1.16e-03 | 8.60e-04 | 1.35e-06 | 0.196    | 7.72e-16  |
| ridge       | 0.582    | 0.681   | -0.100 | 0.036 [0.030, 0.041]    | yes | 9.93e-04 | 8.00e-04 | 9.86e-07 | 0.180    | 5.17e-04 | 1.16e-03 | 8.60e-04 | 1.35e-06 | 0.196    | -6.92e-17 |
| lasso       | 0.581    | 0.679   | -0.098 | 0.033 [0.027, 0.039]    | yes | 9.96e-04 | 8.02e-04 | 9.93e-07 | 0.181    | 5.19e-04 | 1.16e-03 | 8.61e-04 | 1.35e-06 | 0.195    | 4.96e-19  |
| elasticnet  | 0.581    | 0.676   | -0.095 | 0.031 [0.025, 0.036]    | yes | 1.00e-03 | 8.05e-04 | 1.00e-06 | 0.182    | 5.22e-04 | 1.16e-03 | 8.61e-04 | 1.35e-06 | 0.195    | 4.51e-19  |
| huber       | 0.575    | 0.717   | -0.142 | 0.071 [0.068, 0.075]    | yes | 9.35e-04 | 7.38e-04 | 8.75e-07 | 0.166    | 4.04e-04 | 1.17e-03 | 8.48e-04 | 1.37e-06 | 0.204    | -1.31e-04 |
| poly2       | 0.606    | 0.668   | -0.062 | 0.022 [0.012, 0.032]    | yes | 1.01e-03 | 8.23e-04 | 1.03e-06 | 0.186    | 5.85e-04 | 1.13e-03 | 8.27e-04 | 1.27e-06 | 0.181    | 1.73e-14  |
| poly3       | 0.610    | 0.654   | -0.044 | 0.008 [0.001, 0.016]    | yes | 1.03e-03 | 8.42e-04 | 1.07e-06 | 0.212    | 6.11e-04 | 1.12e-03 | 8.20e-04 | 1.26e-06 | 0.184    | 8.85e-13  |
| knn         | 0.651    | 0.626   | 0.025  | -0.019 [-0.033, -0.009] | no  | 1.08e-03 | 8.85e-04 | 1.16e-06 | 0.196    | 6.51e-04 | 1.06e-03 | 7.61e-04 | 1.13e-06 | 0.153    | -1.32e-05 |
| svr ★       | 0.640    | 0.717   | -0.077 | 0.072 [0.062, 0.082]    | yes | 9.35e-04 | 7.51e-04 | 8.74e-07 | 0.161    | 4.63e-04 | 1.08e-03 | 7.53e-04 | 1.16e-06 | 0.166    | -1.47e-04 |
| kernelridge | 0.642    | 0.660   | -0.018 | 0.015 [-0.000, 0.027]   | no  | 1.02e-03 | 8.43e-04 | 1.05e-06 | 0.231    | 6.11e-04 | 1.07e-03 | 7.70e-04 | 1.16e-06 | 0.208    | -1.05e-06 |
| gbm         | 0.613    | 0.600   | 0.013  | -0.046 [-0.055, -0.037] | no  | 1.11e-03 | 9.23e-04 | 1.24e-06 | 0.217    | 6.91e-04 | 1.12e-03 | 8.15e-04 | 1.25e-06 | 0.166    | -7.09e-08 |
| rf          | 0.692    | 0.581   | 0.111  | -0.064 [-0.088, -0.044] | no  | 1.14e-03 | 9.46e-04 | 1.29e-06 | 0.218    | 7.35e-04 | 9.97e-04 | 7.14e-04 | 9.95e-07 | 0.134    | 4.54e-07  |
| extratrees  | 0.682    | 0.607   | 0.075  | -0.038 [-0.057, -0.024] | no  | 1.10e-03 | 9.17e-04 | 1.21e-06 | 0.209    | 7.00e-04 | 1.01e-03 | 7.25e-04 | 1.03e-06 | 0.136    | 6.46e-20  |
| mlp         | 0.656    | 0.644   | 0.013  | -0.002 [-0.020, 0.014]  | no  | 1.05e-03 | 8.70e-04 | 1.10e-06 | 0.191    | 6.50e-04 | 1.05e-03 | 7.55e-04 | 1.11e-06 | 0.151    | -3.14e-06 |
| gp          | 0.631    | 0.605   | 0.026  | -0.040 [-0.057, -0.027] | no  | 1.10e-03 | 9.09e-04 | 1.22e-06 | 0.204    | 6.87e-04 | 1.09e-03 | 7.85e-04 | 1.19e-06 | 0.158    | 5.93e-06  |

best-by-test-R2 = **svr** (0.717) · linear 0.681 · persistence 0.646 · H=1800s · 1787.8s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

AI

Sigma=60s | horizon=30m | metric=AI

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.578    | 0.675   | -0.098 | 0.030 [0.022, 0.036]    | yes | 1.00e-03 | 8.06e-04 | 1.00e-06 | 0.183    | 5.24e-04 | 1.17e-03 | 8.66e-04 | 1.36e-06 | 0.199    | 7.54e-16  |
| ridge       | 0.578    | 0.675   | -0.098 | 0.030 [0.022, 0.036]    | yes | 1.00e-03 | 8.06e-04 | 1.00e-06 | 0.183    | 5.24e-04 | 1.17e-03 | 8.66e-04 | 1.36e-06 | 0.199    | -7.04e-17 |
| lasso       | 0.577    | 0.673   | -0.095 | 0.027 [0.020, 0.034]    | yes | 1.01e-03 | 8.10e-04 | 1.01e-06 | 0.184    | 5.29e-04 | 1.17e-03 | 8.66e-04 | 1.36e-06 | 0.198    | 4.96e-19  |
| elasticnet  | 0.577    | 0.670   | -0.093 | 0.024 [0.017, 0.031]    | yes | 1.01e-03 | 8.14e-04 | 1.02e-06 | 0.185    | 5.34e-04 | 1.17e-03 | 8.66e-04 | 1.37e-06 | 0.198    | 4.54e-19  |
| huber ★     | 0.571    | 0.710   | -0.139 | 0.065 [0.061, 0.069]    | yes | 9.46e-04 | 7.46e-04 | 8.96e-07 | 0.170    | 4.07e-04 | 1.18e-03 | 8.54e-04 | 1.38e-06 | 0.208    | -1.31e-04 |
| poly2       | 0.600    | 0.654   | -0.054 | 0.008 [-0.003, 0.018]   | no  | 1.04e-03 | 8.40e-04 | 1.07e-06 | 0.191    | 6.01e-04 | 1.14e-03 | 8.34e-04 | 1.29e-06 | 0.182    | 1.94e-14  |
| poly3       | 0.611    | 0.645   | -0.033 | -0.001 [-0.014, 0.011]  | no  | 1.05e-03 | 8.61e-04 | 1.10e-06 | 0.217    | 6.35e-04 | 1.12e-03 | 8.16e-04 | 1.25e-06 | 0.197    | 1.30e-12  |
| knn         | 0.648    | 0.614   | 0.034  | -0.031 [-0.047, -0.018] | no  | 1.09e-03 | 9.02e-04 | 1.19e-06 | 0.201    | 6.75e-04 | 1.07e-03 | 7.65e-04 | 1.14e-06 | 0.154    | -1.53e-05 |
| svr         | 0.637    | 0.708   | -0.071 | 0.063 [0.052, 0.072]    | yes | 9.50e-04 | 7.65e-04 | 9.03e-07 | 0.165    | 4.79e-04 | 1.08e-03 | 7.57e-04 | 1.17e-06 | 0.166    | -1.48e-04 |
| kernelridge | 0.641    | 0.645   | -0.004 | -0.001 [-0.018, 0.012]  | no  | 1.05e-03 | 8.64e-04 | 1.10e-06 | 0.284    | 6.33e-04 | 1.08e-03 | 7.72e-04 | 1.16e-06 | 0.235    | 5.83e-09  |
| gbm         | 0.610    | 0.597   | 0.013  | -0.048 [-0.059, -0.039] | no  | 1.12e-03 | 9.28e-04 | 1.24e-06 | 0.218    | 6.98e-04 | 1.12e-03 | 8.19e-04 | 1.26e-06 | 0.167    | -4.28e-08 |
| rf          | 0.689    | 0.559   | 0.130  | -0.087 [-0.112, -0.063] | no  | 1.17e-03 | 9.70e-04 | 1.36e-06 | 0.225    | 7.62e-04 | 1.00e-03 | 7.18e-04 | 1.00e-06 | 0.134    | 5.36e-07  |

| learner    | train R2 | test R2 | gap   | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias  |
|------------|----------|---------|-------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| extratrees | 0.680    | 0.587   | 0.093 | -0.058 [-0.079, -0.040] | no  | 1.13e-03 | 9.43e-04 | 1.28e-06 | 0.217    | 7.34e-04 | 1.02e-03 | 7.28e-04 | 1.03e-06 | 0.136    | 1.54e-20 |
| mlp        | 0.654    | 0.631   | 0.023 | -0.015 [-0.036, 0.004]  | no  | 1.07e-03 | 8.87e-04 | 1.14e-06 | 0.196    | 6.71e-04 | 1.06e-03 | 7.59e-04 | 1.12e-06 | 0.151    | 3.88e-06 |
| gp         | 0.633    | 0.606   | 0.027 | -0.039 [-0.055, -0.026] | no  | 1.10e-03 | 9.13e-04 | 1.22e-06 | 0.205    | 6.96e-04 | 1.09e-03 | 7.84e-04 | 1.19e-06 | 0.158    | 7.66e-06 |

best-by-test-R2 = **huber** (0.710) · linear 0.675 · persistence 0.646 · H=1800s · 1904.1s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

horizon=1hSigma=60s · horizon=1h (H=3600s) — metrics: LE + AI

LESigma=60s | horizon=1h | metric=LE

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.510    | 0.627   | -0.118 | 0.032 [0.025, 0.038]    | yes | 1.48e-03 | 1.20e-03 | 2.19e-06 | 0.192    | 8.21e-04 | 1.71e-03 | 1.25e-03 | 2.91e-06 | 0.205    | 8.50e-16  |
| ridge       | 0.510    | 0.627   | -0.118 | 0.032 [0.025, 0.038]    | yes | 1.48e-03 | 1.20e-03 | 2.19e-06 | 0.192    | 8.21e-04 | 1.71e-03 | 1.25e-03 | 2.91e-06 | 0.205    | 5.49e-17  |
| lasso       | 0.510    | 0.626   | -0.116 | 0.030 [0.023, 0.036]    | yes | 1.49e-03 | 1.21e-03 | 2.21e-06 | 0.192    | 8.23e-04 | 1.71e-03 | 1.25e-03 | 2.91e-06 | 0.205    | -2.94e-19 |
| elasticnet  | 0.510    | 0.624   | -0.114 | 0.028 [0.021, 0.034]    | yes | 1.49e-03 | 1.21e-03 | 2.22e-06 | 0.193    | 8.25e-04 | 1.71e-03 | 1.25e-03 | 2.92e-06 | 0.205    | -4.88e-19 |
| huber ★     | 0.500    | 0.683   | -0.183 | 0.087 [0.083, 0.092]    | yes | 1.37e-03 | 1.08e-03 | 1.87e-06 | 0.171    | 6.16e-04 | 1.72e-03 | 1.23e-03 | 2.97e-06 | 0.217    | -2.24e-04 |
| poly2       | 0.534    | 0.598   | -0.064 | 0.003 [-0.011, 0.015]   | no  | 1.54e-03 | 1.25e-03 | 2.37e-06 | 0.203    | 9.19e-04 | 1.66e-03 | 1.21e-03 | 2.77e-06 | 0.195    | 3.02e-14  |
| poly3       | 0.541    | 0.577   | -0.037 | -0.018 [-0.027, -0.009] | no  | 1.58e-03 | 1.29e-03 | 2.49e-06 | 0.225    | 9.67e-04 | 1.65e-03 | 1.20e-03 | 2.73e-06 | 0.204    | 3.20e-13  |
| knn         | 0.587    | 0.550   | 0.037  | -0.045 [-0.061, -0.033] | no  | 1.63e-03 | 1.34e-03 | 2.65e-06 | 0.213    | 1.01e-03 | 1.57e-03 | 1.12e-03 | 2.45e-06 | 0.165    | -1.73e-05 |
| svr         | 0.569    | 0.682   | -0.113 | 0.087 [0.074, 0.098]    | yes | 1.37e-03 | 1.10e-03 | 1.87e-06 | 0.167    | 6.69e-04 | 1.60e-03 | 1.10e-03 | 2.56e-06 | 0.184    | -2.74e-04 |
| kernelridge | 0.577    | 0.593   | -0.016 | -0.003 [-0.018, 0.010]  | no  | 1.55e-03 | 1.28e-03 | 2.40e-06 | 0.290    | 9.36e-04 | 1.59e-03 | 1.13e-03 | 2.51e-06 | 0.266    | -6.47e-06 |
| gbm         | 0.559    | 0.541   | 0.018  | -0.054 [-0.066, -0.043] | no  | 1.64e-03 | 1.37e-03 | 2.70e-06 | 0.225    | 1.05e-03 | 1.62e-03 | 1.17e-03 | 2.62e-06 | 0.172    | -1.84e-07 |
| rf          | 0.635    | 0.488   | 0.147  | -0.107 [-0.137, -0.082] | no  | 1.74e-03 | 1.45e-03 | 3.01e-06 | 0.238    | 1.15e-03 | 1.47e-03 | 1.04e-03 | 2.17e-06 | 0.142    | 6.20e-08  |
| extratrees  | 0.625    | 0.527   | 0.098  | -0.068 [-0.089, -0.052] | no  | 1.67e-03 | 1.40e-03 | 2.78e-06 | 0.225    | 1.08e-03 | 1.49e-03 | 1.06e-03 | 2.23e-06 | 0.144    | -9.69e-21 |
| mlp         | 0.592    | 0.540   | 0.052  | -0.055 [-0.080, -0.034] | no  | 1.65e-03 | 1.38e-03 | 2.71e-06 | 0.219    | 1.08e-03 | 1.56e-03 | 1.12e-03 | 2.43e-06 | 0.161    | 2.77e-05  |
| gp          | 0.566    | 0.587   | -0.022 | -0.008 [-0.023, 0.004]  | no  | 1.56e-03 | 1.28e-03 | 2.43e-06 | 0.201    | 9.50e-04 | 1.61e-03 | 1.14e-03 | 2.58e-06 | 0.172    | -3.12e-05 |

best-by-test-R2 = **huber** (0.683) · linear 0.627 · persistence 0.596 · H=3600s · 1740.9s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

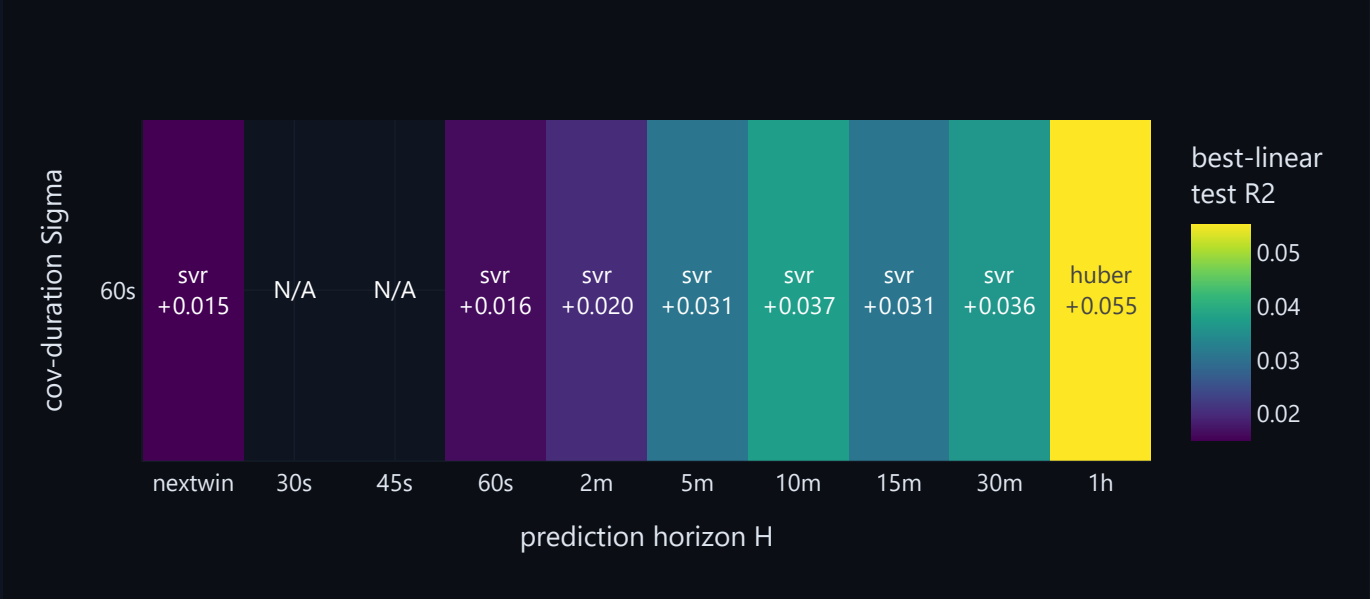
AISigma=60s | horizon=1h | metric=AI

| learner     | train R2 | test R2 | gap    | dR2 [CI95]              | sig | te.RMSE  | te.MAE   | te.MSE   | te.QLIKE | te.bias  | tr.RMSE  | tr.MAE   | tr.MSE   | tr.QLIKE | tr.bias   |
|-------------|----------|---------|--------|-------------------------|-----|----------|----------|----------|----------|----------|----------|----------|----------|----------|-----------|
| linear      | 0.506    | 0.620   | -0.113 | 0.024 [0.015, 0.033]    | yes | 1.50e-03 | 1.22e-03 | 2.24e-06 | 0.195    | 8.35e-04 | 1.71e-03 | 1.26e-03 | 2.93e-06 | 0.207    | -1.83e-15 |
| ridge       | 0.506    | 0.620   | -0.113 | 0.024 [0.015, 0.033]    | yes | 1.50e-03 | 1.22e-03 | 2.24e-06 | 0.195    | 8.35e-04 | 1.71e-03 | 1.26e-03 | 2.93e-06 | 0.207    | 5.59e-17  |
| lasso       | 0.506    | 0.618   | -0.111 | 0.022 [0.013, 0.031]    | yes | 1.50e-03 | 1.22e-03 | 2.25e-06 | 0.196    | 8.39e-04 | 1.71e-03 | 1.26e-03 | 2.94e-06 | 0.207    | -1.97e-19 |
| elasticnet  | 0.506    | 0.615   | -0.109 | 0.020 [0.011, 0.028]    | yes | 1.51e-03 | 1.22e-03 | 2.27e-06 | 0.196    | 8.44e-04 | 1.71e-03 | 1.26e-03 | 2.94e-06 | 0.207    | -4.59e-19 |
| huber ★     | 0.497    | 0.675   | -0.179 | 0.079 [0.075, 0.085]    | yes | 1.38e-03 | 1.09e-03 | 1.91e-06 | 0.175    | 6.25e-04 | 1.73e-03 | 1.24e-03 | 2.99e-06 | 0.220    | -2.24e-04 |
| poly2       | 0.528    | 0.582   | -0.054 | -0.014 [-0.030, -0.000] | no  | 1.57e-03 | 1.28e-03 | 2.46e-06 | 0.207    | 9.44e-04 | 1.67e-03 | 1.22e-03 | 2.80e-06 | 0.195    | -2.96e-14 |
| poly3       | 0.544    | 0.569   | -0.025 | -0.027 [-0.044, -0.011] | no  | 1.59e-03 | 1.32e-03 | 2.54e-06 | 0.230    | 1.01e-03 | 1.65e-03 | 1.19e-03 | 2.71e-06 | 0.208    | -3.04e-13 |
| knn         | 0.586    | 0.533   | 0.053  | -0.063 [-0.082, -0.046] | no  | 1.66e-03 | 1.38e-03 | 2.75e-06 | 0.219    | 1.05e-03 | 1.57e-03 | 1.12e-03 | 2.46e-06 | 0.165    | -2.08e-05 |
| svr         | 0.568    | 0.659   | -0.091 | 0.063 [0.050, 0.075]    | yes | 1.42e-03 | 1.14e-03 | 2.01e-06 | 0.176    | 7.24e-04 | 1.60e-03 | 1.10e-03 | 2.57e-06 | 0.183    | -2.71e-04 |
| kernelridge | 0.577    | 0.578   | -0.001 | -0.018 [-0.036, -0.003] | no  | 1.58e-03 | 1.31e-03 | 2.49e-06 | 0.296    | 9.76e-04 | 1.59e-03 | 1.13e-03 | 2.52e-06 | 0.270    | -5.61e-06 |
| gbm         | 0.558    | 0.540   | 0.018  | -0.056 [-0.070, -0.045] | no  | 1.65e-03 | 1.38e-03 | 2.71e-06 | 0.226    | 1.06e-03 | 1.62e-03 | 1.17e-03 | 2.63e-06 | 0.172    | -2.47e-07 |
| rf          | 0.634    | 0.452   | 0.182  | -0.143 [-0.178, -0.113] | no  | 1.80e-03 | 1.50e-03 | 3.23e-06 | 0.249    | 1.21e-03 | 1.47e-03 | 1.05e-03 | 2.17e-06 | 0.142    | 9.98e-08  |
| extratrees  | 0.624    | 0.501   | 0.123  | -0.095 [-0.123, -0.071] | no  | 1.71e-03 | 1.44e-03 | 2.94e-06 | 0.235    | 1.14e-03 | 1.50e-03 | 1.06e-03 | 2.24e-06 | 0.144    | 3.23e-21  |
| mlp         | 0.591    | 0.517   | 0.073  | -0.078 [-0.110, -0.052] | no  | 1.69e-03 | 1.40e-03 | 2.84e-06 | 0.225    | 1.10e-03 | 1.56e-03 | 1.11e-03 | 2.43e-06 | 0.163    | -1.47e-05 |
| gp          | 0.566    | 0.579   | -0.013 | -0.017 [-0.034, -0.003] | no  | 1.58e-03 | 1.30e-03 | 2.48e-06 | 0.205    | 9.77e-04 | 1.61e-03 | 1.14e-03 | 2.58e-06 | 0.171    | -3.04e-05 |

best-by-test-R2 = **huber** (0.675) · linear 0.620 · persistence 0.596 · H=3600s · 1764.1s. te.\* = OOS test losses; tr.\* = in-sample train losses (RMSE/MAE/bias in vol units, MSE=vol^2, QLIKE on variance).

**Capacity - does a flexible learner beat linear?** Per cell we compare the BEST learner's test R2 to the LINEAR learner's test R2. The heatmap shows the capacity gain (best - linear test R2) over (horizon x cov-duration), one panel per metric; the table lists the best learner per cell. A near-0 gain means the predictive map is essentially linear at that horizon; a positive gain means capacity helps.

Sigma=60sCapacity gain (best - linear test R2) [metric=LE]LE



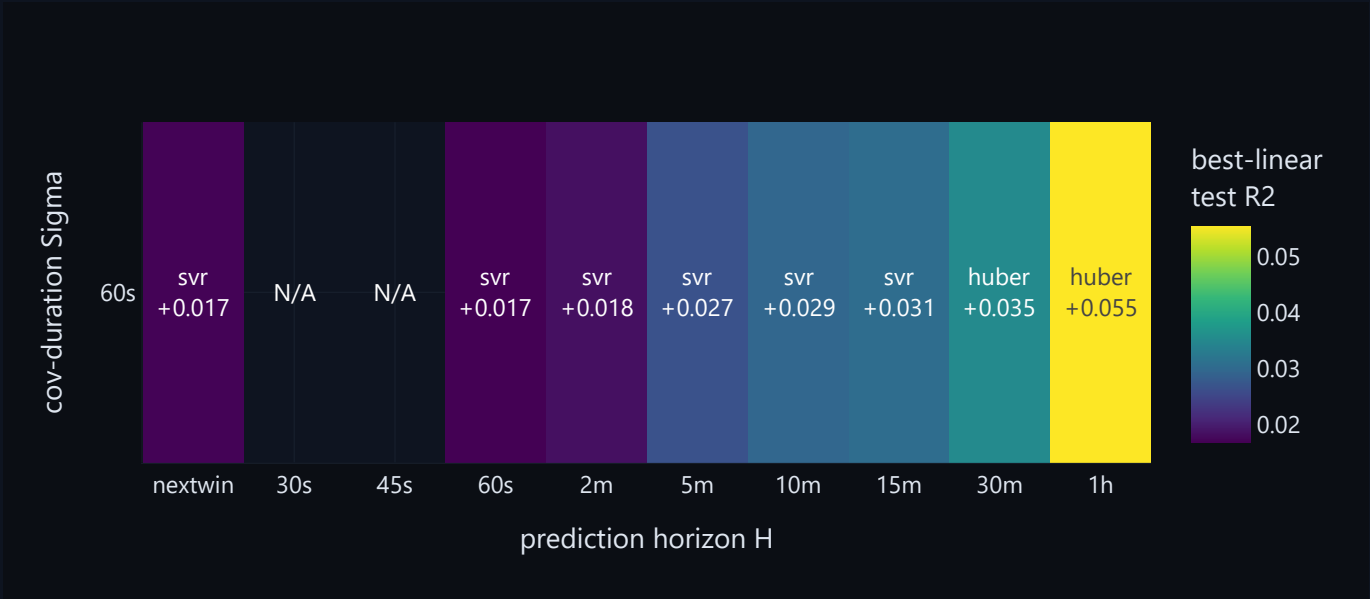
**In plain terms** How much a flexible learner beats plain linear at each cov-dur x horizon.

**Math / formula** gain = R2\_OOS(best-by-test) - R2\_OOS(linear); per (Sigma, H).

**How to read it** cell text = best learner + gain; bright/high = capacity helps, ~0 = linear is enough.

**Insight** metric LE; empty cells = not finished yet.

Sigma=60sCapacity gain (best - linear test R2) [metric=AI]AI



**In plain terms** How much a flexible learner beats plain linear at each cov-dur x horizon.

**Math / formula** gain = R2\_OOS(best-by-test) - R2\_OOS(linear); per (Sigma, H).

**How to read it** cell text = best learner + gain; bright/high = capacity helps, ~0 = linear is enough.

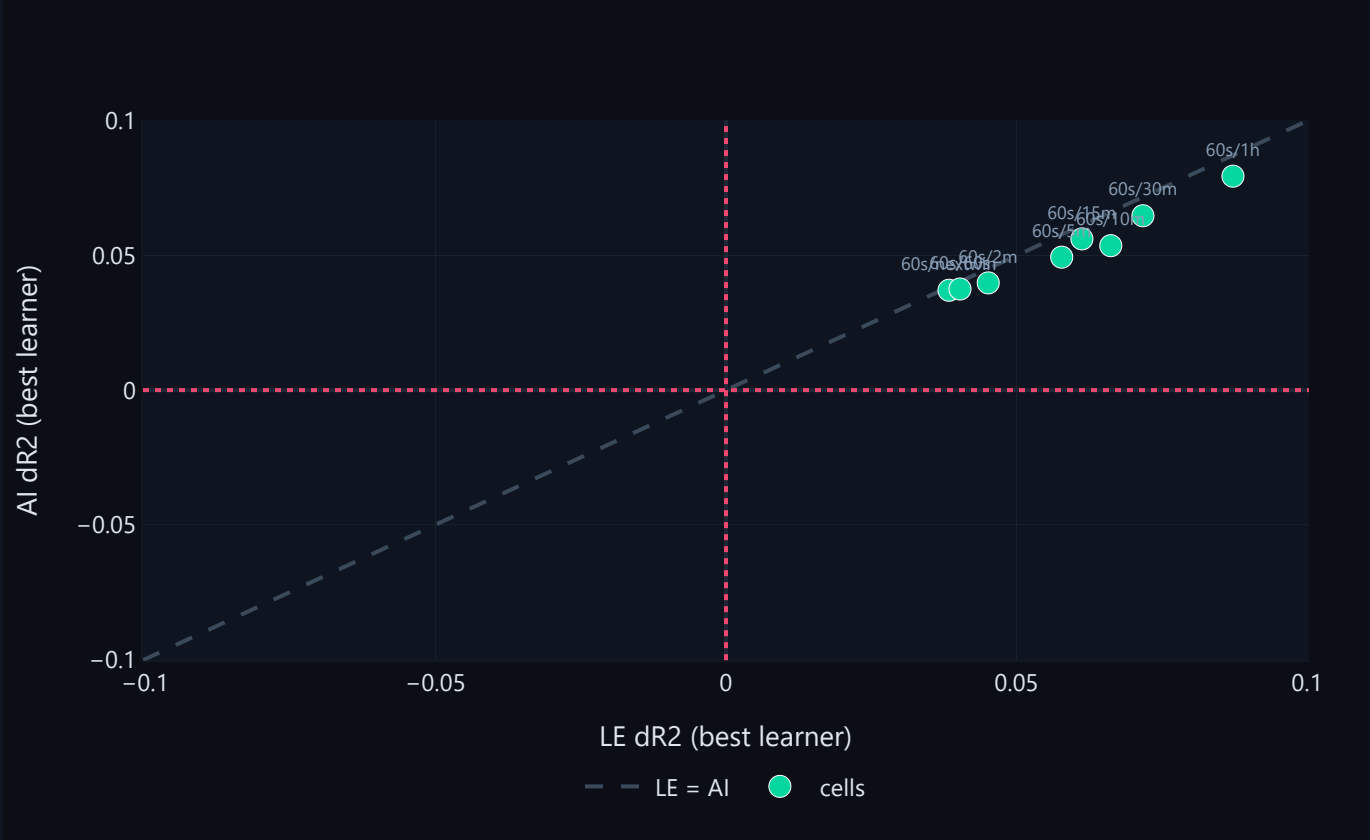
**Insight** metric AI; empty cells = not finished yet.

Best learner + capacity per cell (every finished cov-dur x horizon x metric).

| Sigma | horizon | metric | best learner | best test R2 | linear test R2 | capacity gain | persistence R2 | best overfit gap |
|-------|---------|--------|--------------|--------------|----------------|---------------|----------------|------------------|
| 60s   | nextwin | LE     | svr          | 0.626        | 0.610          | 0.015         | 0.587          | -0.046           |
| 60s   | nextwin | AI     | svr          | 0.624        | 0.607          | 0.017         | 0.587          | -0.047           |
| 60s   | 60s     | LE     | svr          | 0.628        | 0.612          | 0.016         | 0.588          | -0.045           |
| 60s   | 60s     | AI     | svr          | 0.625        | 0.608          | 0.017         | 0.588          | -0.045           |
| 60s   | 2m      | LE     | svr          | 0.676        | 0.656          | 0.020         | 0.631          | -0.054           |
| 60s   | 2m      | AI     | svr          | 0.671        | 0.652          | 0.018         | 0.631          | -0.052           |
| 60s   | 5m      | LE     | svr          | 0.733        | 0.703          | 0.031         | 0.676          | -0.056           |
| 60s   | 5m      | AI     | svr          | 0.725        | 0.698          | 0.027         | 0.676          | -0.051           |
| 60s   | 10m     | LE     | svr          | 0.748        | 0.711          | 0.037         | 0.682          | -0.061           |
| 60s   | 10m     | AI     | svr          | 0.736        | 0.706          | 0.029         | 0.682          | -0.051           |
| 60s   | 15m     | LE     | svr          | 0.737        | 0.707          | 0.031         | 0.676          | -0.058           |
| 60s   | 15m     | AI     | svr          | 0.732        | 0.702          | 0.031         | 0.676          | -0.055           |
| 60s   | 30m     | LE     | svr          | 0.717        | 0.681          | 0.036         | 0.646          | -0.077           |
| 60s   | 30m     | AI     | huber        | 0.710        | 0.675          | 0.035         | 0.646          | -0.139           |
| 60s   | 1h      | LE     | huber        | 0.683        | 0.627          | 0.055         | 0.596          | -0.183           |
| 60s   | 1h      | AI     | huber        | 0.675        | 0.620          | 0.055         | 0.596          | -0.179           |

**LE vs AI - is the answer metric-robust?** **AI** **LE** For each (cov-dur x horizon) with BOTH metrics finished, do LE and AI agree on (a) the best learner, and (b) the SIGN of the dial's dR2 over persistence? AI is the headline / LE the diagnostic (Step-6 convention). **metric\_robust** = same dR2 sign AND both significant-or-both-not.

**Sigma=60s** LE vs AI dial edge (best-learner dR2)



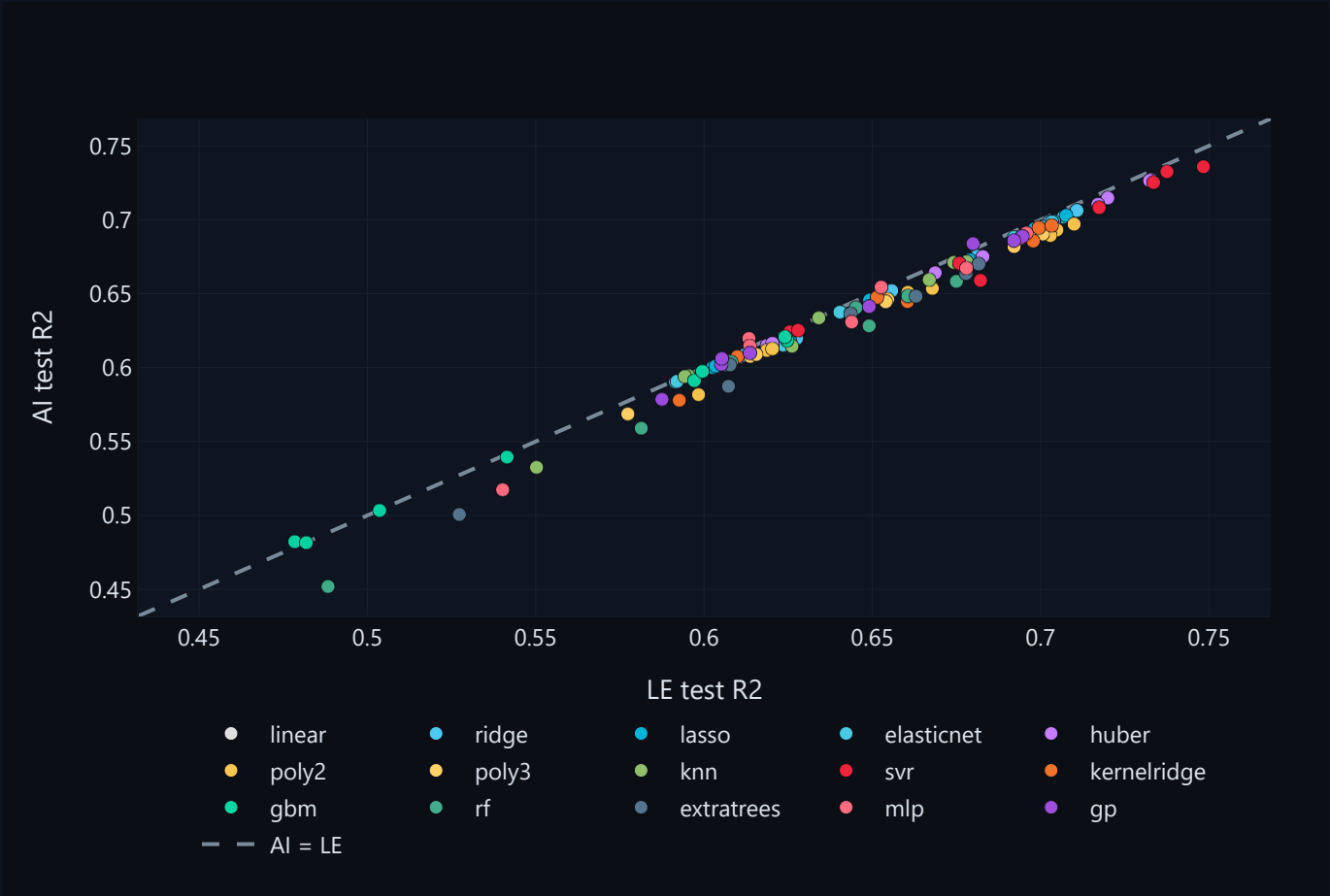
**In plain terms** Do the two SPD rulers (Log-Euclidean vs affine-invariant) agree on the dial's OOS edge?

**Math / formula** point = one (Sigma, H) cell; x = LE dR2, y = AI dR2; the diagonal is perfect agreement.

**How to read it** points on/near the diagonal and in the same quadrant = metric-robust; off-diagonal = fragile.

**Insight** clustering along y=x and in the ++ quadrant would confirm the AI dial reproduces the LE verdict.

**Sigma=60s** LE vs AI test-R2 per (cov-dur x horizon x learner) **AI** **LE**



**In plain terms** Every finished (cov-dur x horizon x learner) cell plotted as LE vs AI test R2.

**Math / formula** point = one (Sigma, H, model); x = LE test R2, y = AI test R2; dashed = AI=LE.

**How to read it** tight clustering on the diagonal = the SPD metric barely changes accuracy (metric-robust).

**Insight** color = learner; each spans all cov-dur x horizon cells where both metrics finished.

**LE-vs-AI agreement per (cov-dur x horizon).** same best learner? same dR2 sign? metric-robust?

| Sigma | horizon | best LE | best AI | same best | dR2 LE | dR2 AI | same sign | metric-robust |
|-------|---------|---------|---------|-----------|--------|--------|-----------|---------------|
| 60s   | nextwin | svr     | svr     | True      | 0.038  | 0.037  | True      | True          |
| 60s   | 60s     | svr     | svr     | True      | 0.040  | 0.038  | True      | True          |
| 60s   | 2m      | svr     | svr     | True      | 0.045  | 0.040  | True      | True          |
| 60s   | 5m      | svr     | svr     | True      | 0.058  | 0.049  | True      | True          |
| 60s   | 10m     | svr     | svr     | True      | 0.066  | 0.054  | True      | True          |
| 60s   | 15m     | svr     | svr     | True      | 0.061  | 0.056  | True      | True          |
| 60s   | 30m     | svr     | huber   | False     | 0.072  | 0.065  | True      | True          |
| 60s   | 1h      | huber   | huber   | True      | 0.087  | 0.079  | True      | True          |

**Complete Step-7 performance picture.** The full-matrix tab renders every learner; here we render the per-CELL summary + the per-YEAR breakdown of the best learner (train/test R2, dR2 over persistence, QLIKE, RMSE) for every finished (cov-dur x horizon x metric). Every row is labelled with cov-duration + horizon + metric. QLIKE is on variance; RMSE/MAE/bias in vol units.

Best-learner loss battery per cell (train R2 / test R2 / gap / dR2[CI] / sig / RMSE / MAE / QLIKE / bias / n\_test).

| Sigma | horizon | metric | best learner | train R2 | test R2 | gap    | dR2 [CI95]          | sig | RMSE     | MAE      | QLIKE | bias     | n_test |
|-------|---------|--------|--------------|----------|---------|--------|---------------------|-----|----------|----------|-------|----------|--------|
| 60s   | nextwin | LE     | svr          | 0.579    | 0.626   | -0.046 | 0.038 [0.031,0.045] | yes | 2.49e-04 | 1.94e-04 | 0.375 | 6.55e-05 | 312652 |
| 60s   | nextwin | AI     | svr          | 0.577    | 0.624   | -0.047 | 0.037 [0.032,0.042] | yes | 2.49e-04 | 1.94e-04 | 0.371 | 6.48e-05 | 312652 |
| 60s   | 60s     | LE     | svr          | 0.583    | 0.628   | -0.045 | 0.040 [0.033,0.047] | yes | 2.41e-04 | 1.89e-04 | 0.356 | 6.62e-05 | 298209 |
| 60s   | 60s     | AI     | svr          | 0.580    | 0.625   | -0.045 | 0.038 [0.031,0.044] | yes | 2.41e-04 | 1.90e-04 | 0.356 | 6.62e-05 | 298209 |
| 60s   | 2m      | LE     | svr          | 0.622    | 0.676   | -0.054 | 0.045 [0.040,0.051] | yes | 2.90e-04 | 2.28e-04 | 0.283 | 1.03e-04 | 309044 |
| 60s   | 2m      | AI     | svr          | 0.619    | 0.671   | -0.052 | 0.040 [0.034,0.045] | yes | 2.93e-04 | 2.30e-04 | 0.285 | 1.03e-04 | 309044 |
| 60s   | 5m      | LE     | svr          | 0.677    | 0.733   | -0.056 | 0.058 [0.052,0.065] | yes | 3.90e-04 | 3.07e-04 | 0.200 | 1.57e-04 | 309043 |
| 60s   | 5m      | AI     | svr          | 0.674    | 0.725   | -0.051 | 0.049 [0.042,0.057] | yes | 3.96e-04 | 3.12e-04 | 0.203 | 1.67e-04 | 309043 |
| 60s   | 10m     | LE     | svr          | 0.687    | 0.748   | -0.061 | 0.066 [0.059,0.074] | yes | 5.23e-04 | 4.13e-04 | 0.167 | 2.28e-04 | 305725 |
| 60s   | 10m     | AI     | svr          | 0.685    | 0.736   | -0.051 | 0.054 [0.045,0.061] | yes | 5.36e-04 | 4.26e-04 | 0.173 | 2.49e-04 | 305725 |
| 60s   | 15m     | LE     | svr          | 0.680    | 0.737   | -0.058 | 0.061 [0.052,0.070] | yes | 6.48e-04 | 5.16e-04 | 0.162 | 3.07e-04 | 302315 |
| 60s   | 15m     | AI     | svr          | 0.677    | 0.732   | -0.055 | 0.056 [0.046,0.065] | yes | 6.54e-04 | 5.22e-04 | 0.165 | 3.16e-04 | 302315 |
| 60s   | 30m     | LE     | svr          | 0.640    | 0.717   | -0.077 | 0.072 [0.062,0.082] | yes | 9.35e-04 | 7.51e-04 | 0.161 | 4.63e-04 | 292276 |
| 60s   | 30m     | AI     | huber        | 0.571    | 0.710   | -0.139 | 0.065 [0.061,0.069] | yes | 9.46e-04 | 7.46e-04 | 0.170 | 4.07e-04 | 292276 |
| 60s   | 1h      | LE     | huber        | 0.500    | 0.683   | -0.183 | 0.087 [0.083,0.092] | yes | 1.37e-03 | 1.08e-03 | 0.171 | 6.16e-04 | 271823 |
| 60s   | 1h      | AI     | huber        | 0.497    | 0.675   | -0.179 | 0.079 [0.075,0.085] | yes | 1.38e-03 | 1.09e-03 | 0.175 | 6.25e-04 | 271823 |

Per-year OOS breakdown of the best learner (purged walk-forward test years). train R2 / test R2 / persistence R2 / dR2 / QLIKE / RMSE.

| Sigma | horizon | metric | learner | test year | train R2 | test R2 | persistence R2 | dR2    | QLIKE | RMSE     |
|-------|---------|--------|---------|-----------|----------|---------|----------------|--------|-------|----------|
| 60s   | nextwin | LE     | svr     | 2023      | 0.569    | 0.700   | 0.632          | 0.068  | 0.401 | 2.62e-04 |
| 60s   | nextwin | LE     | svr     | 2024      | 0.590    | 0.565   | 0.522          | 0.044  | 0.378 | 2.30e-04 |
| 60s   | nextwin | LE     | svr     | 2025      | 0.576    | 0.558   | 0.570          | -0.013 | 0.349 | 2.53e-04 |
| 60s   | nextwin | AI     | svr     | 2023      | 0.569    | 0.699   | 0.632          | 0.067  | 0.391 | 2.62e-04 |
| 60s   | nextwin | AI     | svr     | 2024      | 0.587    | 0.556   | 0.522          | 0.034  | 0.370 | 2.32e-04 |
| 60s   | nextwin | AI     | svr     | 2025      | 0.573    | 0.563   | 0.570          | -0.008 | 0.352 | 2.52e-04 |
| 60s   | 60s     | LE     | svr     | 2023      | 0.578    | 0.699   | 0.631          | 0.068  | 0.361 | 2.56e-04 |
| 60s   | 60s     | LE     | svr     | 2024      | 0.595    | 0.555   | 0.518          | 0.037  | 0.359 | 2.25e-04 |
| 60s   | 60s     | LE     | svr     | 2025      | 0.577    | 0.574   | 0.576          | -0.002 | 0.350 | 2.40e-04 |
| 60s   | 60s     | AI     | svr     | 2023      | 0.577    | 0.694   | 0.631          | 0.064  | 0.353 | 2.58e-04 |
| 60s   | 60s     | AI     | svr     | 2024      | 0.591    | 0.547   | 0.518          | 0.029  | 0.360 | 2.27e-04 |
| 60s   | 60s     | AI     | svr     | 2025      | 0.572    | 0.579   | 0.576          | 0.003  | 0.355 | 2.38e-04 |
| 60s   | 2m      | LE     | svr     | 2023      | 0.602    | 0.752   | 0.672          | 0.080  | 0.302 | 3.01e-04 |
| 60s   | 2m      | LE     | svr     | 2024      | 0.633    | 0.605   | 0.570          | 0.034  | 0.265 | 2.72e-04 |
| 60s   | 2m      | LE     | svr     | 2025      | 0.621    | 0.611   | 0.614          | -0.003 | 0.281 | 2.97e-04 |
| 60s   | 2m      | AI     | svr     | 2023      | 0.602    | 0.748   | 0.672          | 0.077  | 0.303 | 3.03e-04 |
| 60s   | 2m      | AI     | svr     | 2024      | 0.631    | 0.588   | 0.570          | 0.018  | 0.269 | 2.78e-04 |
| 60s   | 2m      | AI     | svr     | 2025      | 0.617    | 0.612   | 0.614          | -0.002 | 0.282 | 2.96e-04 |
| 60s   | 5m      | LE     | svr     | 2023      | 0.653    | 0.815   | 0.710          | 0.105  | 0.216 | 3.88e-04 |
| 60s   | 5m      | LE     | svr     | 2024      | 0.691    | 0.662   | 0.626          | 0.036  | 0.182 | 3.71e-04 |
| 60s   | 5m      | LE     | svr     | 2025      | 0.678    | 0.658   | 0.659          | -0.002 | 0.202 | 4.12e-04 |
| 60s   | 5m      | AI     | svr     | 2023      | 0.652    | 0.807   | 0.710          | 0.097  | 0.221 | 3.95e-04 |
| 60s   | 5m      | AI     | svr     | 2024      | 0.689    | 0.640   | 0.626          | 0.013  | 0.188 | 3.83e-04 |
| 60s   | 5m      | AI     | svr     | 2025      | 0.672    | 0.659   | 0.659          | 0.000  | 0.200 | 4.11e-04 |
| 60s   | 10m     | LE     | svr     | 2023      | 0.658    | 0.833   | 0.713          | 0.120  | 0.176 | 5.09e-04 |
| 60s   | 10m     | LE     | svr     | 2024      | 0.702    | 0.670   | 0.639          | 0.031  | 0.153 | 5.04e-04 |
| 60s   | 10m     | LE     | svr     | 2025      | 0.689    | 0.671   | 0.666          | 0.004  | 0.172 | 5.55e-04 |
| 60s   | 10m     | AI     | svr     | 2023      | 0.655    | 0.821   | 0.713          | 0.108  | 0.186 | 5.27e-04 |
| 60s   | 10m     | AI     | svr     | 2024      | 0.701    | 0.639   | 0.639          | 0.000  | 0.161 | 5.28e-04 |
| 60s   | 10m     | AI     | svr     | 2025      | 0.685    | 0.672   | 0.666          | 0.006  | 0.173 | 5.54e-04 |
| 60s   | 15m     | LE     | svr     | 2023      | 0.642    | 0.832   | 0.704          | 0.128  | 0.164 | 6.19e-04 |
| 60s   | 15m     | LE     | svr     | 2024      | 0.698    | 0.628   | 0.636          | -0.008 | 0.155 | 6.49e-04 |
| 60s   | 15m     | LE     | svr     | 2025      | 0.682    | 0.669   | 0.662          | 0.007  | 0.166 | 6.74e-04 |
| 60s   | 15m     | AI     | svr     | 2023      | 0.641    | 0.826   | 0.704          | 0.121  | 0.169 | 6.31e-04 |
| 60s   | 15m     | AI     | svr     | 2024      | 0.696    | 0.618   | 0.636          | -0.018 | 0.159 | 6.57e-04 |
| 60s   | 15m     | AI     | svr     | 2025      | 0.678    | 0.670   | 0.662          | 0.008  | 0.167 | 6.73e-04 |
| 60s   | 30m     | LE     | svr     | 2023      | 0.593    | 0.814   | 0.669          | 0.146  | 0.167 | 9.07e-04 |
| 60s   | 30m     | LE     | svr     | 2024      | 0.661    | 0.605   | 0.611          | -0.007 | 0.153 | 9.27e-04 |

| Sigma | horizon | metric | learner | test year | train R2 | test R2 | persistence R2 | dR2   | QLIKE | RMSE     |
|-------|---------|--------|---------|-----------|----------|---------|----------------|-------|-------|----------|
| 60s   | 30m     | LE     | svr     | 2025      | 0.645    | 0.646   | 0.635          | 0.011 | 0.163 | 9.70e-04 |
| 60s   | 30m     | AI     | huber   | 2023      | 0.528    | 0.723   | 0.669          | 0.055 | 0.220 | 1.11e-03 |
| 60s   | 30m     | AI     | huber   | 2024      | 0.578    | 0.699   | 0.611          | 0.087 | 0.142 | 8.09e-04 |
| 60s   | 30m     | AI     | huber   | 2025      | 0.585    | 0.698   | 0.635          | 0.063 | 0.147 | 8.96e-04 |
| 60s   | 1h      | LE     | huber   | 2023      | 0.436    | 0.699   | 0.614          | 0.085 | 0.225 | 1.60e-03 |
| 60s   | 1h      | LE     | huber   | 2024      | 0.510    | 0.674   | 0.564          | 0.109 | 0.141 | 1.16e-03 |
| 60s   | 1h      | LE     | huber   | 2025      | 0.519    | 0.663   | 0.590          | 0.073 | 0.146 | 1.31e-03 |
| 60s   | 1h      | AI     | huber   | 2023      | 0.436    | 0.682   | 0.614          | 0.068 | 0.234 | 1.64e-03 |
| 60s   | 1h      | AI     | huber   | 2024      | 0.505    | 0.671   | 0.564          | 0.107 | 0.143 | 1.16e-03 |
| 60s   | 1h      | AI     | huber   | 2025      | 0.516    | 0.667   | 0.590          | 0.076 | 0.147 | 1.30e-03 |

GridSearch best-params + wall-time per cell.

| Sigma | horizon | metric | best learner | best_params              | elapsed (s) |
|-------|---------|--------|--------------|--------------------------|-------------|
| 60s   | nextwin | LE     | svr          | {"C": 3.0, "gamma": 0.1} | 1801.4      |
| 60s   | nextwin | AI     | svr          | {"C": 3.0, "gamma": 0.1} | 1840.1      |
| 60s   | 60s     | LE     | svr          | {"C": 3.0, "gamma": 0.1} | 1771.6      |
| 60s   | 60s     | AI     | svr          | {"C": 3.0, "gamma": 0.1} | 1814.1      |
| 60s   | 2m      | LE     | svr          | {"C": 3.0, "gamma": 0.1} | 1780.9      |
| 60s   | 2m      | AI     | svr          | {"C": 3.0, "gamma": 0.1} | 1836.1      |
| 60s   | 5m      | LE     | svr          | {"C": 3.0, "gamma": 0.1} | 1803.2      |
| 60s   | 5m      | AI     | svr          | {"C": 1.0, "gamma": 0.1} | 1850.7      |
| 60s   | 10m     | LE     | svr          | {"C": 3.0, "gamma": 0.1} | 1803.3      |
| 60s   | 10m     | AI     | svr          | {"C": 3.0, "gamma": 0.1} | 1867.5      |
| 60s   | 15m     | LE     | svr          | {"C": 3.0, "gamma": 0.1} | 1777.1      |
| 60s   | 15m     | AI     | svr          | {"C": 3.0, "gamma": 0.1} | 1859.8      |
| 60s   | 30m     | LE     | svr          | {"C": 3.0, "gamma": 0.1} | 1787.8      |
| 60s   | 30m     | AI     | huber        | (no grid / defaults)     | 1904.1      |
| 60s   | 1h      | LE     | huber        | (no grid / defaults)     | 1740.9      |
| 60s   | 1h      | AI     | huber        | (no grid / defaults)     | 1764.1      |

Step 7 (Track A) — horizon x cov-duration bake-off (PARTIAL render: 16 cell(s) finished)

Holding the 30/45/60 s masked covariance state fixed, the fixed Step-6 predictor `[rv, s_slow, s_fast, v]` is baked off with 15 learners against a realized-vol target at growing horizons, under LE and AI features. The headline object is **dR2(horizon)** - the dial's OOS edge over persistence - which is small/coincident at next-window and tests whether it grows (leading) with H. Current peaks: 60s/AI: peak dR2 0.079 @ 1h [sig]. **next-window is COINCIDENT; there is no next-day forecast here (deferred Track B).**

Holistic scoreboard — every finished cov-dur x horizon x metric, color-coded

GREEN = dial adds OOS value / robust / capacity helps / significant

AMBER = small / coincident / borderline / not significant

RED = dial hurts / dR2<0 / metric-fragile

| cell (horizon · metric)       | best learner                                                   | dR2 over persistence [CI95] | sig | best / linear test R2 | capacity gain | QLIKE | LE-vs-AI |
|-------------------------------|----------------------------------------------------------------|-----------------------------|-----|-----------------------|---------------|-------|----------|
| Sigma = 60s covariance window |                                                                |                             |     |                       |               |       |          |
| horizon=nextwin · LE          | svr                                                            | 0.038 [0.031,0.045]         | sig | 0.626 / 0.610         | +0.015        | 0.375 | robust   |
| horizon=nextwin · AI          | svr                                                            | 0.037 [0.032,0.042]         | sig | 0.624 / 0.607         | +0.017        | 0.371 | robust   |
| horizon=30s · LE              | N/A — horizon < cov window (H=30s ≤ Sigma=60s); not measurable |                             |     |                       |               |       | skip     |
| horizon=30s · AI              | N/A — horizon < cov window (H=30s ≤ Sigma=60s); not measurable |                             |     |                       |               |       | skip     |
| horizon=45s · LE              | N/A — horizon < cov window (H=45s ≤ Sigma=60s); not measurable |                             |     |                       |               |       | skip     |
| horizon=45s · AI              | N/A — horizon < cov window (H=45s ≤ Sigma=60s); not measurable |                             |     |                       |               |       | skip     |
| horizon=60s · LE              | svr                                                            | 0.040 [0.033,0.047]         | sig | 0.628 / 0.612         | +0.016        | 0.356 | robust   |
| horizon=60s · AI              | svr                                                            | 0.038 [0.031,0.044]         | sig | 0.625 / 0.608         | +0.017        | 0.356 | robust   |
| horizon=2m · LE               | svr                                                            | 0.045 [0.040,0.051]         | sig | 0.676 / 0.656         | +0.020        | 0.283 | robust   |
| horizon=2m · AI               | svr                                                            | 0.040 [0.034,0.045]         | sig | 0.671 / 0.652         | +0.018        | 0.285 | robust   |
| horizon=5m · LE               | svr                                                            | 0.058 [0.052,0.065]         | sig | 0.733 / 0.703         | +0.031        | 0.200 | robust   |
| horizon=5m · AI               | svr                                                            | 0.049 [0.042,0.057]         | sig | 0.725 / 0.698         | +0.027        | 0.203 | robust   |
| horizon=10m · LE              | svr                                                            | 0.066 [0.059,0.074]         | sig | 0.748 / 0.711         | +0.037        | 0.167 | robust   |
| horizon=10m · AI              | svr                                                            | 0.054 [0.045,0.061]         | sig | 0.736 / 0.706         | +0.029        | 0.173 | robust   |
| horizon=15m · LE              | svr                                                            | 0.061 [0.052,0.070]         | sig | 0.737 / 0.707         | +0.031        | 0.162 | robust   |
| horizon=15m · AI              | svr                                                            | 0.056 [0.046,0.065]         | sig | 0.732 / 0.702         | +0.031        | 0.165 | robust   |
| horizon=30m · LE              | svr                                                            | 0.072 [0.062,0.082]         | sig | 0.717 / 0.681         | +0.036        | 0.161 | robust   |
| horizon=30m · AI              | huber                                                          | 0.065 [0.061,0.069]         | sig | 0.710 / 0.675         | +0.035        | 0.170 | robust   |
| horizon=1h · LE               | huber                                                          | 0.087 [0.083,0.092]         | sig | 0.683 / 0.627         | +0.055        | 0.171 | robust   |
| horizon=1h · AI               | huber                                                          | 0.079 [0.075,0.085]         | sig | 0.675 / 0.620         | +0.055        | 0.175 | robust   |

dR2 GREEN = positive AND 95% CI clears 0 (dial leads persistence here); AMBER = positive but not significant (coincident); RED = negative. Capacity gain = best-learner minus linear test R2. LE-vs-AI shows "pending" until the matching AI cell finishes (AI cells run after all LE cells).

**What this establishes.** Step 7 (Track A) traces the **horizon-of-predictability** curve for the covariance stress dial while holding the covariance state fixed. It converts Step-6's single next-window coincidence check into a full dR2(H) sweep and, for the first time, builds the **AI (affine-invariant) dial** that Steps 5/6 deferred - so the LE-vs-AI robustness claim is tested on a real predictor, not just a detector. The 15-learner bake-off separates "does the dial carry signal" from "does the answer need capacity" at each horizon.

Honest caveats / negatives.

- next-window is COINCIDENT.** At H=nextwin the dial co-moves with vol and persistence dominates; a small dR2 there is expected, not a failure.
- Longer horizons test LEADING, but overlapping targets inflate apparent skill** - the block-bootstrap CI over days (not the point estimate) is the honest significance test.
- No next-day forecast here.** Track B (daily HAR / QLIKE / Diebold-Mariano / MCS) is a separate deferred run; Track A stays intraday (nextwin..1h).
- The covariance state is unchanged** - no new Sigma window is built; this is a horizon sweep on the existing 30/45/60 s masked cloud.
- PARTIAL.** This render reflects only the cells finished so far (16); re-run the builder when the detached grid completes for the full scoreboard.